SF SHOW

OPERATOR'S MANUAL

THE SHOW TEMPLATE, THE E16 SCENES AND THE OXI ONE MKII
ONE LIVE SET TWO E16 SCENES ONE SEQUENCER

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STRANGE FREQUENCIES

E16 PAGES, PRESSES AND ADDRESSES IN THIS MANUAL ARE GENERATED FROM CONTROLLERS/E16/LAYOUT.PY.

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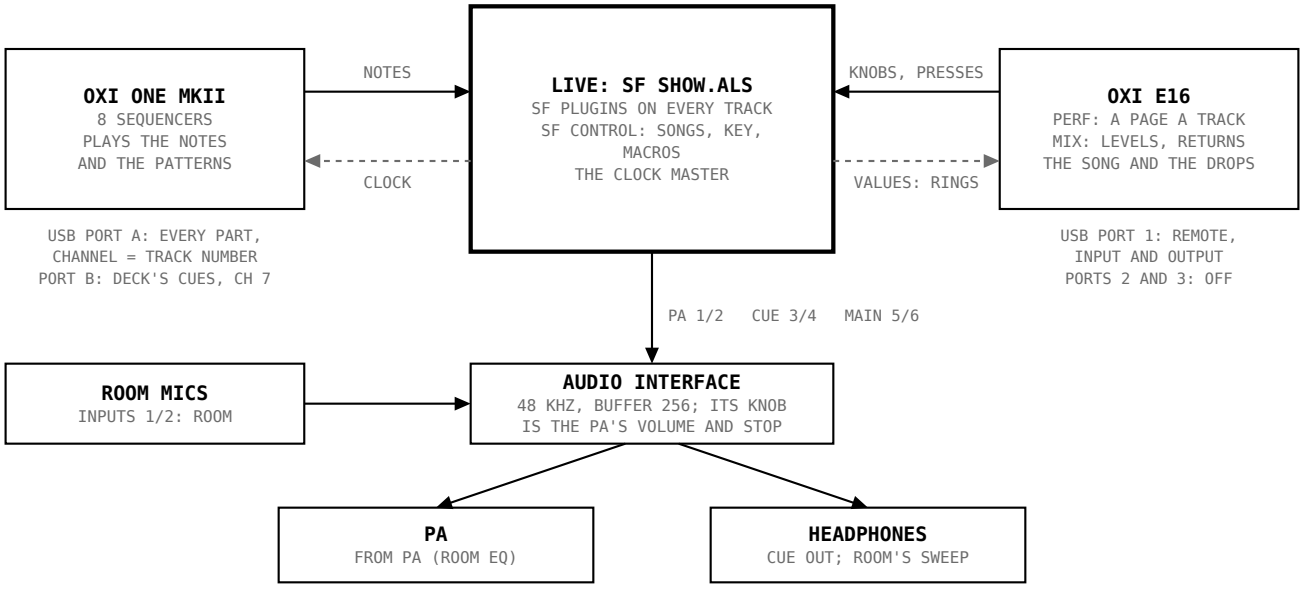
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1 THE RIG

Three devices, three jobs. The OXI One MkII plays the notes and the patterns. Live, running SF SHOW.als, makes the sound. The E16 plays the sound, the balance and the drops. Nothing on stage is loaded, routed or mapped: everything below is set once.



DEVICE	ITS JOB
OXI ONE MKII	Every note, rhythm and pattern change.
LIVE: SF SHOW.ALS	Every sound (one SF plugin per role), the songs, the master chain, the recording, the clock.
OXI E16	A page per track, every ENERGY on one page, the song call and the drop (PERF); the levels, the returns and the macros (MIX).
AUDIO INTERFACE	Its output knob is the PA's volume and the emergency stop: the room goes quiet, the recording carries on.

ONE NUMBER PER TRACK

— **A track's number is everything:** 3 OPEN HAT is the third track in Live, plays on channel 3 of the ONE's port A, and is page 3 of the E16 scene PERF, its knobs on channel 3 of the E16. Live, the ONE and the E16 read the same way.

— **On the E16 a track goes by a code** of three letters: KCK, CHH, OHH, CLP, PRC, BAS, RMB, LED, CRD, PAD, VOX, and the groups DRM, LOW, MUS. Its page is titled with it, its ENERGY knob reads KCK NRG, its volume KCK VOL.

— **The order:** 1 KICK, the drums (2 CLOSED HAT, 3 OPEN HAT, 4 CLAP, 5 PERC), the low end (6 BASS, 7 RUMBLE), the music (8 LEAD, 9 CHORD, 10 PAD, 11 VOICE).

— **A track is a role, not an instrument.** Put another instrument on 9 CHORD and it keeps its name, its channel and its page: map the page to the new instrument (Cmd-M; turn to the page before you click a parameter, since the page tells Live its track as it comes up) and nothing else moves.

THE TWO HANDS

— **NRG, page 12:** every track's ENERGY and every group's, the song call and the build. The whole arrangement under one hand.

— **A track's page:** its instrument, deep. Knobs 15 and 16 are NEXT SONG and BUILD on every page, so the song and the drop are always under the same finger.

WHAT NEVER CHANGES IN A SHOW

— The tracks, the routing, the E16 scenes and the mappings. A song is a stored set of values in every SF plugin, recalled on the bar: nothing is loaded between tracks.

2 THE SHOW TEMPLATE

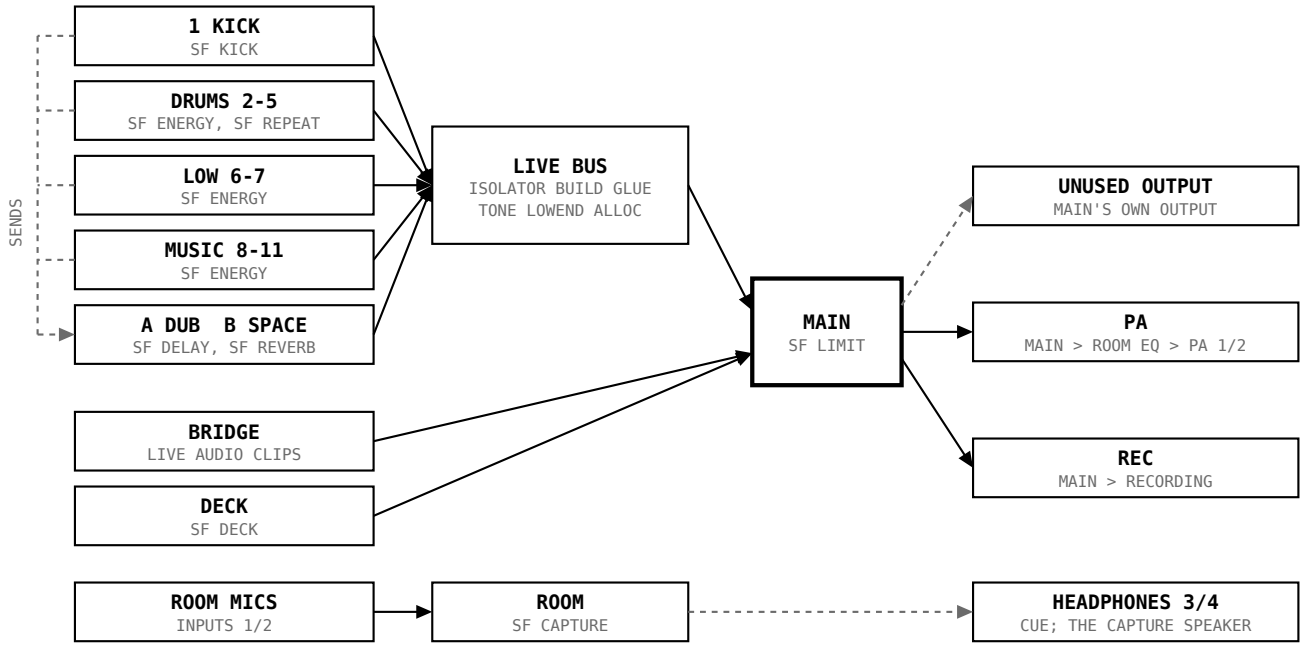
SF SHOW.als is built: every track, group and return, every SF plugin with its starting settings, the routing, every instrument's MIDI From on its OXI One port and channel, and every address of both E16 scenes mapped. Set Live's own settings once (below), open it, and keep it as the template for every show.

TRACKS

TRACK	DEVICES	MIDI FROM	OUT
CONTROL	SF CONTROL	ONE port A, ch 16	Main
1 KICK	SF KICK; SF ENERGY	ONE port A, ch 1	LIVE BUS
DRUMS (group)	SF ENERGY, PROFILE DRUMS; SF REPEAT		LIVE BUS
2 CLOSED HAT	SF PERC: closed hat; SF ENERGY	port A, ch 2	DRUMS
3 OPEN HAT	SF PERC: open hat or ride; SF ENERGY	port A, ch 3	DRUMS
4 CLAP	SF PERC: clap or rim; SF ENERGY	port A, ch 4	DRUMS
5 PERC	SF STRIKE; SF ENERGY	port A, ch 5	DRUMS
LOW (group)	SF ENERGY, PROFILE BASS		LIVE BUS
6 BASS	SF BASS; SF ENERGY	port A, ch 6	LOW
7 RUMBLE	SF RUMBLE, SOURCE KICK BUS; SF ENERGY	none: it follows the kick	LOW
MUSIC (group)	SF ENERGY, PROFILE MUSIC		LIVE BUS
8 LEAD	SF LEAD: ACID 303-TYPE; SF ENERGY	port A, ch 8	MUSIC
9 CHORD	SF LEAD: STAB DUB; SF ENERGY	port A, ch 9	MUSIC
10 PAD	SF PAD: GRANULAR; SF ENERGY	port A, ch 10	MUSIC
11 VOICE	SF VOICE; SF ENERGY	port A, ch 11	MUSIC
BRIDGE	Live audio clips		Main
DECK	SF DECK	port B, ch 7 (hot cues; optional)	Main
LIVE BUS	SF ISOLATOR > BUILD > GLUE > TONE > LOWEND; SF ALLOC (off at first)	sends disabled	Main
PA	SF ROOM EQ > SF LIMIT (safety)	audio: Main, Post Mixer	the PA outputs (1/2)
REC	nothing	audio: Main, Post Mixer; Monitor Off	Sends Only
ROOM	SF CAPTURE	audio: Ext. In 1/2, the room mics; Monitor In	the headphones (3/4)
A DUB (return)	SF DELAY, DRY off		LIVE BUS
B SPACE (return)	SF REVERB, DRY off		LIVE BUS
Main	SF LIMIT		5/6: a pair that goes nowhere

Every instrument track is on Monitor In: it plays from the ONE whether or not it is armed, and ends in SF ENERGY (its knob on NRG). MIDI From is set on every instrument: in Live the ONE's port A is OXI ONE MKII (Port 1), port B is (Port 2). The kick stays outside DRUMS and LOW, so the drum group's REPEAT never takes it and the bass and the rumble fade together while the kick holds the floor.

ROUTING



— Main is the set, after SF LIMIT: exporting it gives the release mix, and its own output goes nowhere. PA and REC both take the Main after its fader. The venue's correction (ROOM EQ) exists only on PA, so the recording never hears it.

— PA's SF LIMIT is a safety: CEILING -0.5 dB, no clip, AUTO off. The set's peaks are already at -1 dBTP, so it only catches what the room EQ's phase pushes past. Keep PA's fader at 0 dB: the PA's level is the interface's.

— Never send LIVE BUS to PA: BRIDGE, DECK and SF LIMIT would be left out of the PA, and REC would record nothing of the set.

— **Every send on LIVE BUS is disabled.** The returns feed LIVE BUS, and Live can't compensate a return's delay on a track whose matching send is active, even at zero.

— Pull LIVE BUS down and the live instruments and their effects go while BRIDGE and DECK play on. Turn the interface's output down and the room goes quiet while REC keeps recording.

— ROOM is the room mics, a stereo pair. Armed, it records them with the set. Its SF CAPTURE never passes them (the set never hears the mics): it only plays SF TERROIR's sweep, to a speaker on the headphone out, when you capture the room (chapter 8).

LIVE'S OWN SETTINGS, ONCE PER COMPUTER

PORT (SETTINGS > TEMPO & MIDI)	TRACK	SYNC	REMOTE
OXI ONE MKII (Port 1) and (Port 2), input	on	off	off
OXI ONE MKII (Port 1), output	off	on	off
OXI E16 (Port 1), input and output	off	off	on
every other port	off	off	off

— Takeover Mode: Value Scaling. Audio: 48 kHz, buffer 256. These live in Live's preferences, not in any set: nothing in SF SHOW.als can set them.

— **MIDI Clock Sync Delay** on the ONE's Port 1 output (its row unfolded with the triangle at its left): **0**. The ONE's notes come back to Live a little late (the clock goes out, the notes come back, buffers both

ways: on Nov 7, 40 ms), and nothing needs them on the beat. BUILD lands its drop on the kick it hears (LAND ON KICK; its KICK readout shows how late the kick reaches it), and Pocket and the effects that cut on the beat (GATE, STRIP, ENERGY's chop, DUCK's GRID, REPEAT, DECK) follow where SF KICK's kicks land. On Nov 7, -70 ms there moved the kick not at all.

— **TRACK FOCUS (optional):** copy the folder SF_E16 (controllers/live) into ~/Music/Ableton/User Library/Remote Scripts and restart Live; then a free Control Surface slot: **SF E16**, Input OXI E16 (Port 1), Output None. A PERF page then selects its track (chapter 4). Settings > Plug-Ins: Multiple Plug-In Windows and **Auto-Hide Plug-In Windows** on, and open each instrument's window once: from then on a page brings up its track's windows.

— Before Live 12.4 the tab is called Link, Tempo & MIDI.

OPEN IT, CHECK, SAVE

1. **MIDI From** is set on every instrument, as in the table above (chapter 3 shows the ONE's side). If your Live names the ONE differently, a track shows its port as missing: set it by the table. Never All Ins: every instrument would play every part.
2. **Inputs and outputs**, set for the PA on 1/2 and headphones on 3/4. PA to the pair that feeds the PA (1/2 in the set). Cue Out, on the Main track, to the headphones (3/4), and ROOM there too. Main to a pair that goes nowhere (5/6). Check all three before any sound reaches a PA: the PA pair carries PA and nothing else. The room mics on inputs 1 and 2 (1: the mic on your left as you face the speaker), and in Live's Settings > Audio > Input Config, 1/2 as a stereo pair: ROOM's input.
3. **Two options:** Delay Compensation on; Reduced Latency When Monitoring off.
4. **Save** as a template (File > Save Live Set As Template) or into your show folder. MIDI From and the mappings belong to the set, so every set made from it starts ready.

THE MASTER CHAIN

— Start GLUE, TONE and LOWEND gently: the set doesn't need every processor working all the time.

— Soundcheck the busiest passage of the night and set SF LIMIT's INPUT for 1-3 dB of limiting there. LIMIT AUTO on, AUTO MODE DOWN ONLY, for shows: it holds the set at TARGET and holds through breakdowns.

— SF ALLOC starts off. Turn it on band by band once the arrangement is known, and judge it by ear.

3 THE OXI ONE MKII

The ONE plays every part. Eight sequencers, one per role; every part plays on port A, on its Live track's number. Live is the clock: the ONE follows it.

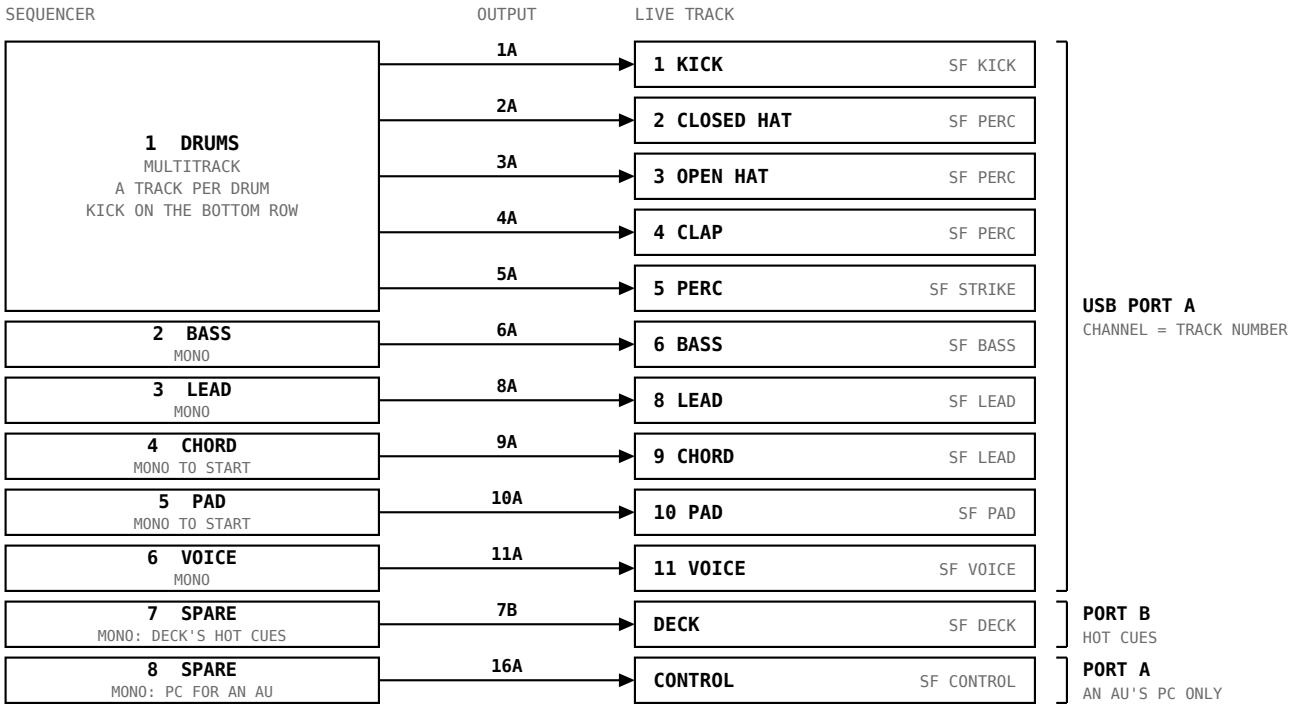
WHAT THE ONE BRINGS

	OXI ONE MKII
SEQUENCERS	8, each in one of six modes: Mono, Poly (up to 7 voices), Chord (up to 8 voices), Multitrack (8 tracks), Stochastic, Matriceal.
PATTERNS	Up to 128 steps; 128 patterns per project.
USB	USB-C for MIDI and power: 32 MIDI channels, as two ports in Live (banks A and B, 16 channels each).
UNUSED HERE	TRS MIDI in and out, Bluetooth MIDI, 8 CV and 8 gate outputs, clock in and out. The rig runs on USB alone.

Sources: Thomann, OXI One MkII; Milk Audio, OXI ONE MkII specifications; Elektronauts: banks A and B appear as two USB MIDI devices. Check menus and limits against your firmware's manual.

SEQUENCERS, OUTPUTS, TRACKS

On the ONE an output is a channel and a port: 1A is channel 1 on port A, 7B channel 7 on port B. Every part plays on port A on its track's number (3 OPEN HAT on 3A); each Live track's MIDI From listens to the same port and channel. SF START has every sequencer set so (controllers/oxi-one).



— **The multitrack's rows,** bottom up: 8 KICK 1A, 7 PERC 5A, 6 CLAP 4A, 5 OPEN HAT 3A, 4 CLOSED HAT 2A; 3-1 off. The kick stays on the bottom row.

— **KICK lane on C1.** With KEY TRACK on, any other note transposes the kick (C1 = TUNE), so C1 is right either way. PERC plays any note.

— **7 RUMBLE needs no lane:** SOURCE KICK BUS follows SF KICK. Channel 7 on port A stays free.

— **9 CHORD:** if the ONE sends whole chords (Chord or Poly mode), set CHORD on the plugin to NONE; if it sends single notes, CHORD voices them. Not both.

— **8 LEAD** slides when notes overlap, over 60 ms like a 303: write legato notes for slides.

— **11 VOICE:** C3 plays slice 1 of the loaded phrase, C#3 slice 2, and so on; below C3 the whole phrase.

— **DECK** takes hot cues on port B, channel 7 (C1-D#1 deck A, C2-D#2 deck B): spare sequencer 7 is set to 7B in SF START.

CLOCK AND PORTS

— **Live is the clock master.** On the ONE: sync USB (stopped, Shift + Stop until the header shows USB, not INT); in its config (Shift + Back), Start Stop Msgs Ignore No, MIDI Transport Msgs Send Off, MIDI Thru off, Auto Sync on (it starts with Live). Never send clock both ways.

— **Live sends its clock only while it plays.** Start and stop from Live, never with the ONE's Play: started on its own, the ONE runs at its own tempo and ignores Live's.

— **Audio:** 48 kHz, buffer 256 (Pocket needs 512 or less), launch quantization 1 bar, record quantization off.

— **The ONE plays a little after Live's beat,** and nothing needs it on the beat: BUILD lands its drop on the kick it hears (its KICK readout shows how late), Pocket and the effects that cut on the beat follow SF KICK. MIDI Clock Sync Delay stays at 0. Leave POCKET OFFSET on SF CONTROL at 0 unless the pump sounds off.

LIVE MIDI PORT	TRACK	SYNC	REMOTE
OXI ONE MKII (Port 1), in: the ONE's port A	on	off	off
OXI ONE MKII (Port 2), in: its port B	on	off	off
OXI ONE MKII (Port 1), out: the clock	off	on	off
OXI E16 (Port 1), in and out	off	off	on
Every other port, E16 Ports 2 and 3 included	off	off	off

PROJECTS AND PATTERNS

— **One ONE project per track** (per piece of music), with a pattern per section: SPARSE, MAIN, VARIATION, EXIT.

— **Every project starts as a copy of SF START**, the startup project: every sequencer set as above, and four scenes of simple patterns in columns 1-4, SPARSE, MAIN, VARIATION, EXIT (controllers/oxi-one: SFSTART.oxipro and its card). Save, an empty slot, Duplic proj here. check_project.py checks the copy.

— **The song comes from the E16, not the project** (chapter 6): any song with any project's patterns, and changing projects never changes the sound. Program Change stays off on every sequencer (it reaches only an AU of SF CONTROL).

— **When patterns change is set on the ONE** (its launch and arranger settings): Live's launch quantization doesn't reach it.

— **KEY on SF CONTROL starts OFF:** the ONE's notes set the pitch, and KICK and RUMBLE are tuned per song with TUNE. Turn KEY on for songs that should change key without reprogramming the ONE.

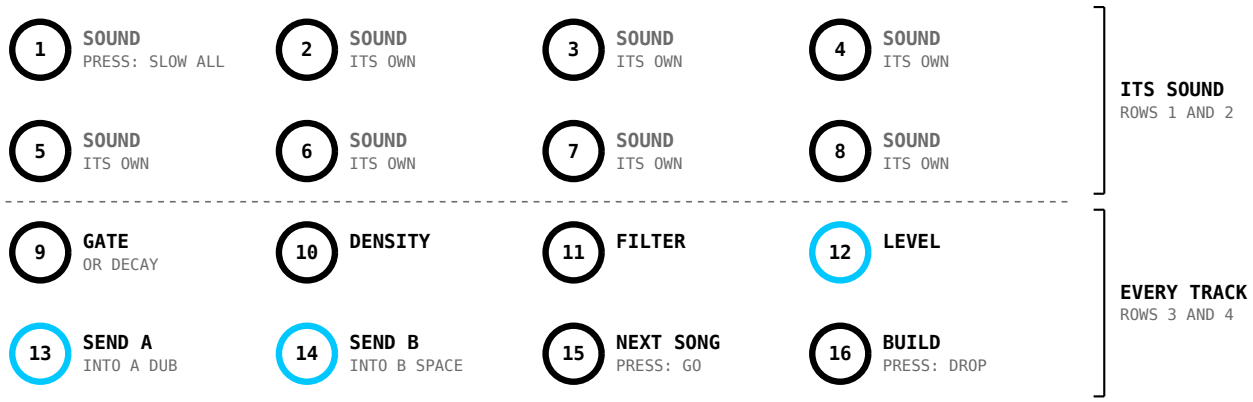
— **Taking the kick out:** mute its lane on the ONE. Pulling the kick's volume down leaves it playing on the bus, so Pocket and RUMBLE keep pumping to a silent kick (which can be the point).

4 THE E16 SCENES

Two E16 scenes run the show. PERF: a page per track (page n is track n, titled with its code) and NRG, every ENERGY knob with the song call and the build. MIX: the levels, the returns and the macros. The set comes with every address of both mapped.

THE GRID

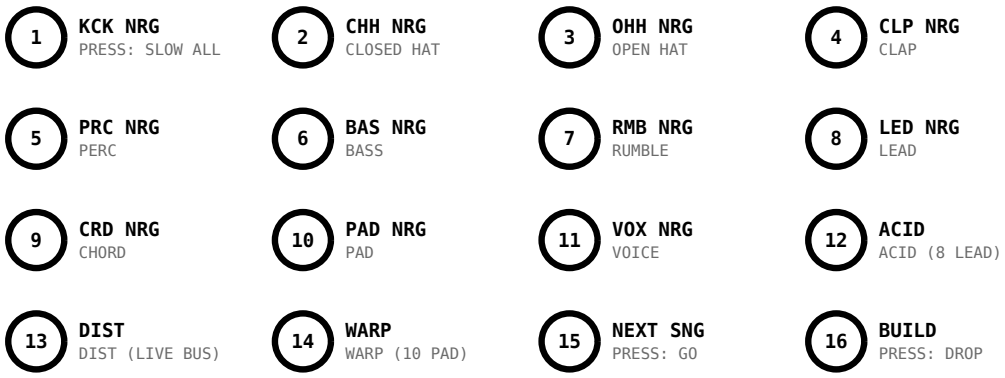
The E16 is a 4 × 4 grid of endless encoders, numbered row by row from the top left. Its screen shows each knob's short name. On PERF, a track's page:



LIGHT BLUE RING: A LEVEL OR A SEND. 15 AND 16 ARE THE SAME KNOBS ON EVERY PERF PAGE.

- **1-8, its sound:** the instrument's own knobs (chapter 5 has every page).
- **9-12, on every track:** GATE (DECAY where the instrument has none), DENSITY, FILTER, LEVEL. The hand learns one layout.
- **13-14, its sends** into A DUB and B SPACE, post-fader, mapped up to 0 dB.
- **15, NEXT SONG:** turn it to pick the next song: nothing changes. **Press it: GO**, the song lands on the next bar line (chapter 6).
- **16, BUILD:** turn it up over a build. **Press it: DROP**, on every page: the music stops on the next sixteenth and the tails ring; press again and the drop comes in on the next kick (chapter 7).
- **Press 1: SLOW ALL.** A quick press flips knob 1 to SLOW ALL on SF CONTROL (the ring turns green): turn it to set how long your next moves take, OFF (at once) to 128 BAR. Press again: the page's own knob 1. Knobs already travelling keep going.

NRG, page 12:



KNOB N IS TRACK N'S SF ENERGY: A PRESS SETS IT BACK TO THE CENTRE (KNOB 1: SLOW ALL). 12-14 ACID, DIST, WARP.

— **The ENERGY knobs:** centre is neutral, left takes energy out, right puts it in (filter, drive, space, echo and chop together). Each track's own knob shapes that part. KCK NRG at zero: no kick. **12 ACID** pushes the 303 on 8 LEAD, **13 DIST** drives the whole mix, **14 WARP** smears the PAD; each is off at zero.

— **MIX:** LEVELS has the volumes in NRG's places (knob n is track n's, 12-14 the groups', 15 DUB and 16 SPACE the returns'); SPACE the two returns, their throws and freezes; MACROS SF CONTROL's six macros, SLOW ALL and the drum roll. Its other nine pages are empty: their knobs send nothing.

RINGS, PAGES, PRESSES

— **Rings, by role** (the OXI App's colour numbers, as set in PERF on Nov 8): white (34) an instrument's own sound, knobs 1-8; red (63) a track's GATE or DECAY, DENSITY and FILTER (9-11) and its NRG knob; light blue (18) a level: LEVEL, the sends, every volume; pink (50) NRG's ACID, DIST and WARP; violet (9) NEXT SONG and BUILD; green (90) knob 1 on SLOW ALL, or a knob travelling (knob 1 itself never shows a glide: it swaps, so it always shows the value of the side it is on). Live sends every value back, so the rings follow song recalls and mouse moves.

— **A recalled knob is yours the moment you turn it.** With Takeover Mode Value Scaling, a knob that disagrees with Live (after a song recall) moves the sound smoothly from where it is, never jumps.

— **A PERF track page selects its track in Live** as it comes up (TRACK FOCUS: with the control surface SF E16 installed, controllers/live/SF_E16): page 6, BAS, selects 6 BASS, as a click on its name would. With Auto-Hide Plug-In Windows on, its open plugin windows come up with it and the others hide: the page you play is the window you see. NRG and MIX's pages leave the selection where it is. Nothing else a page or scene change sends. A knob on several pages (NEXT SONG, BUILD, SLOW ALL) sends the same address from all: several ways to one mapping.

— **A hold press** is a CC, 127 on press and 0 on release: the parameter is on for as long as the knob is down (GO, DROP, THROW, ROLL). **A toggle press** is a note: Live flips the switch on every Note On (FREEZE), so the E16 can never think FREEZE is on while Live has it off. **A centre press** sends 64 on its knob's own address: an ENERGY back to neutral.

— Every press is one the set plays through if it happens by accident (a throw, a roll, a freeze, a drop, an ENERGY back to neutral, a GO with no new song picked). Nothing that would stop the set is on the E16: that is the interface's output knob.

— **On screen, the same grid.** Every SF plugin's window starts with its page of the scenes: its own encoders lit where they sit on the E16, in these colours, the rest dark, its presses under their knobs. Several windows side by side read like the E16 itself. LEAD, PERC and ENERGY are on several tracks: their windows find their encoders by the track's name (VST3).

WHERE	KNOB	PRESS	MESSAGE	LIVE TARGET
every PERF page	1	SLOW ALL, swap	turn 2: CC 12 ch 16	CONTROL > SF CONTROL > SLOW ALL
every PERF page	15	GO, hold	CC 42 ch 16	CONTROL > SF CONTROL > GO
every PERF page	16	DROP, hold	CC 43 ch 15	LIVE BUS > SF BUILD > DROP
NRG	2	ENERGY, centre	CC 28 ch 2, value 64	2 CLOSED HAT > SF ENERGY > ENERGY
NRG	3	ENERGY, centre	CC 28 ch 3, value 64	3 OPEN HAT > SF ENERGY > ENERGY
NRG	4	ENERGY, centre	CC 28 ch 4, value 64	4 CLAP > SF ENERGY > ENERGY
NRG	5	ENERGY, centre	CC 28 ch 5, value 64	5 PERC > SF ENERGY > ENERGY
NRG	6	ENERGY, centre	CC 28 ch 6, value 64	6 BASS > SF ENERGY > ENERGY
NRG	7	ENERGY, centre	CC 28 ch 7, value 64	7 RUMBLE > SF ENERGY > ENERGY
NRG	8	ENERGY, centre	CC 28 ch 8, value 64	8 LEAD > SF ENERGY > ENERGY
NRG	9	ENERGY, centre	CC 28 ch 9, value 64	9 CHORD > SF ENERGY > ENERGY
NRG	10	ENERGY, centre	CC 28 ch 10, value 64	10 PAD > SF ENERGY > ENERGY
NRG	11	ENERGY, centre	CC 28 ch 11, value 64	11 VOICE > SF ENERGY > ENERGY
SPACE	1	THROW, hold	CC 60 ch 12	A DUB > SF DELAY > THROW
SPACE	2	FREEZE, toggle	note 61 (C#3) ch 12	A DUB > SF DELAY > FREEZE
SPACE	5	THROW, hold	CC 64 ch 13	B SPACE > SF REVERB > THROW
SPACE	6	FREEZE, toggle	note 65 (F3) ch 13	B SPACE > SF REVERB > FREEZE
MACROS	9	ROLL, hold	CC 68 ch 14	DRUMS > SF REPEAT > ROLL

ADDRESSES

— **Channel = track:** 1-11 the instrument tracks, 12 A DUB, 13 B SPACE, 14 the groups (DRUMS, LOW, MUSIC and DRUMS' REPEAT), 15 LIVE BUS, 16 CONTROL. Nothing else uses a track's channel.

— **CC = the knob's place + 11 in PERF** (a track's page: CC 12-23), **+ 43 in MIX**; a press its place + 27 in PERF, + 59 in MIX. A track's ENERGY: CC 28 (the groups on 14: 28, 29, 30).

— **Mixer:** CC 102 volume, 103 send A, 104 send B on the track's own channel; channel 14: 102-104 the DRUMS, LOW and MUSIC volumes. CC 6, 38 and 96-101 (data entry and RPN/NRPN) are never used.

— Every address is in the appendix and in controllers/e16/sf-show-map.csv.

LOAD THE SCENES

1. OXI App > Scenes. If no computer folder is assigned yet, assign one.
2. Copy **PERF.oxie16** and **MIX.oxie16** (controllers/e16 in the repository) into the Scenes folder inside it and click Refresh Views. They replace SF SHOW and SF MIX.
3. Drag PERF, then MIX, from On Computer onto two free slots under On Device. The presses come with them.
4. **Auto Transmit off**, for both scenes and in the global settings: sending stored values on a page change would pull Live's faders to whatever the E16 last held. MIDI Thru off.
5. In Live: E16 Port 1 Remote on, input and output; Ports 2 and 3 off (Port 1 is the one the E16 sends on; a second Remote port could take a message twice, and a FREEZE would flip twice). Takeover Mode: Value Scaling.

CHECK IT ONCE

1. Every knob: turn it, the right parameter moves. Every press: the right switch. A track's page moves that track only.
2. Hold DROP, a THROW and ROLL for three seconds each: on for the whole hold, off on release. (A long press also opens the E16's own functions; if one interferes, make that press a toggle.)
3. Shared knobs: on page 1 turn NEXT SONG fully left; on page 6 turn it one step right. The pick moves one step, not jumps. If it jumps, the E16 keeps a value per page: controllers/e16 README, check 4, has the relative fallback.
4. Change pages, change to MIX and back: Live's MIDI input indicator stays dark.
5. Press knob 1 (green) and turn: SLOW ALL moves, the page's knob 1 doesn't. Press again: the page's own.
6. Turn NEXT SONG and press it: the song lands on the bar, the rings follow, and a recalled knob moves smoothly from the new value.

5 THE PAGES

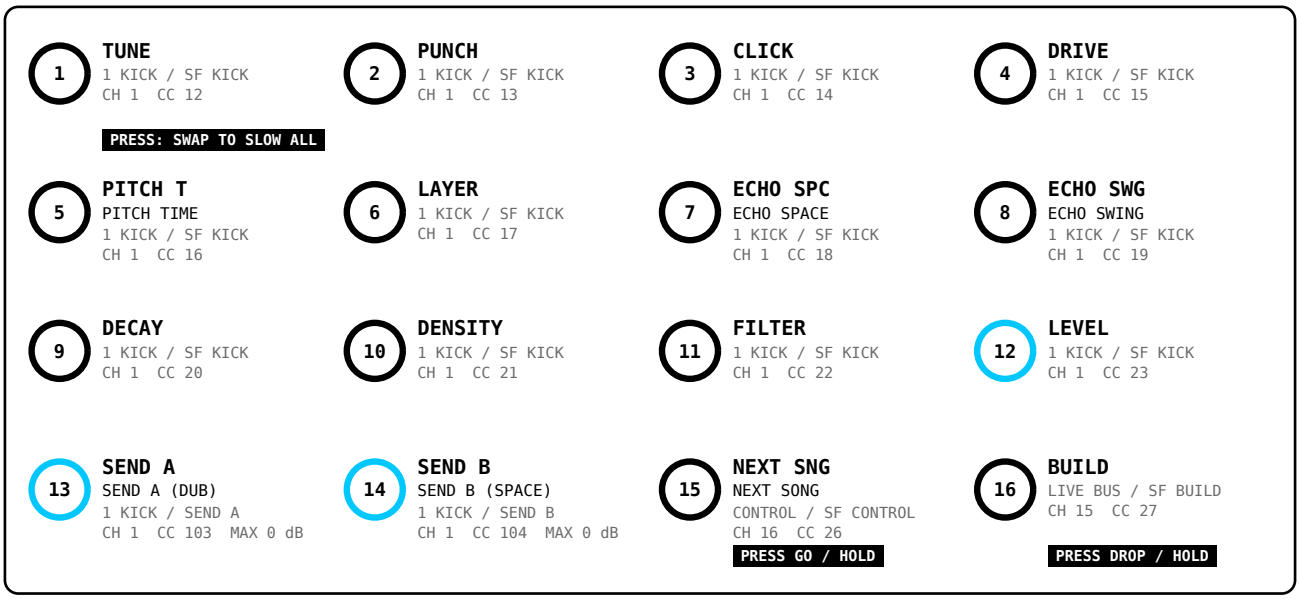
All fifteen pages: PERF's twelve (a page per track, then NRG) and MIX's three. Under each knob: the name on the E16's screen, the parameter, where it lands in Live, its address.

PAGE	KNOBS 1-8	KNOBS 9-16
PERF 1: KCK	TUNE / PUNCH / CLICK / DRIVE / PITCH T / LAYER / ECHO SPC / ECHO SWG	DECAY / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD
PERF 2: CHH	TUNE / TONE / ACCENT / DRIVE / ECHO SPC / ECHO SWG / ECHO PRB / SPREAD	DECAY / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD
PERF 3: OHH	TUNE / TONE / ACCENT / DRIVE / ECHO SPC / ECHO SWG / ECHO PRB / SPREAD	DECAY / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD
PERF 4: CLP	TUNE / TONE / ACCENT / DRIVE / ECHO SPC / ECHO SWG / ECHO PRB / SPREAD	DECAY / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD
PERF 5: PRC	POWER / DISTANCE / SIZE / MATERIAL / POSITION / TENSION / SPREAD / TUNE	DECAY / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD
PERF 6: BAS	CUTOFF / RES / ENV AMT / DECAY / DRIVE / ACCENT / GLIDE / PCKT DEP	GATE / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD
PERF 7: RMB	SIZE / DARK / GRIT / PUMP / TUNE / PREDELAY / WIDTH / PCKT SHP	GATE / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD
PERF 8: LED	CUTOFF / RES / ENV AMT / DECAY / DRIVE / ACCENT / GLIDE / COLOR	GATE / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD
PERF 9: CRD	CUTOFF / RES / ENV AMT / DECAY / ATTACK / SPACE / DETUNE / PCKT DEP	GATE / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD
PERF 10: PAD	CUTOFF / RES / COLOR / DETUNE / ATTACK / DECAY / SPACE / PCKT DEP	GATE / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD
PERF 11: VOX	SIGNAL / TONE / TUNE / DRIVE / ECHO / ECHO TME / ECHO FBK / PCKT DEP	GATE / DENSITY / FILTER / LEVEL / SEND A / SEND B / NEXT SNG / BUILD
PERF 12: NRG	KCK NRG / CHH NRG / OHH NRG / CLP NRG / PRC NRG / BAS NRG / RMB NRG / LED NRG	CRD NRG / PAD NRG / VOX NRG / ACID / DIST / WARP / NEXT SNG / BUILD
MIX 1: LEVELS	KCK VOL / CHH VOL / OHH VOL / CLP VOL / PRC VOL / BAS VOL / RMB VOL / LED VOL	CRD VOL / PAD VOL / VOX VOL / DRM VOL / LOW VOL / MUS VOL / DUB RET / SPC RET
MIX 2: SPACE	DUB TIME / DUB FDBK / DUB TONE / DUB DRV / SPC SIZE / SPC DCY / SPC DARK / SPC DUCK	- / - / - / - / - / - / - / -
MIX 3: MACROS	MACRO 1 / MACRO 2 / MACRO 3 / MACRO 4 / MACRO 5 / MACRO 6 / SLOW ALL / -	RPT LEN / RPT PTCH / - / - / - / - / - / -

On each page below: the drawing, what each knob does (the plugin's own description), and how to play it. On every track page 15 is NEXT SONG (press: GO) and 16 BUILD (press: DROP), and knob 1's press swaps it to SLOW ALL.

PERF 1 KCK

PERF: SF KICK ON 1 KICK, CHANNEL 1



RING: BLACK = A PARAMETER, LIGHT BLUE = A LEVEL. CH / CC: THE ADDRESS.

SF KICK on 1 KICK. TUNE and DECAY are the body, PUNCH and CLICK the attack.

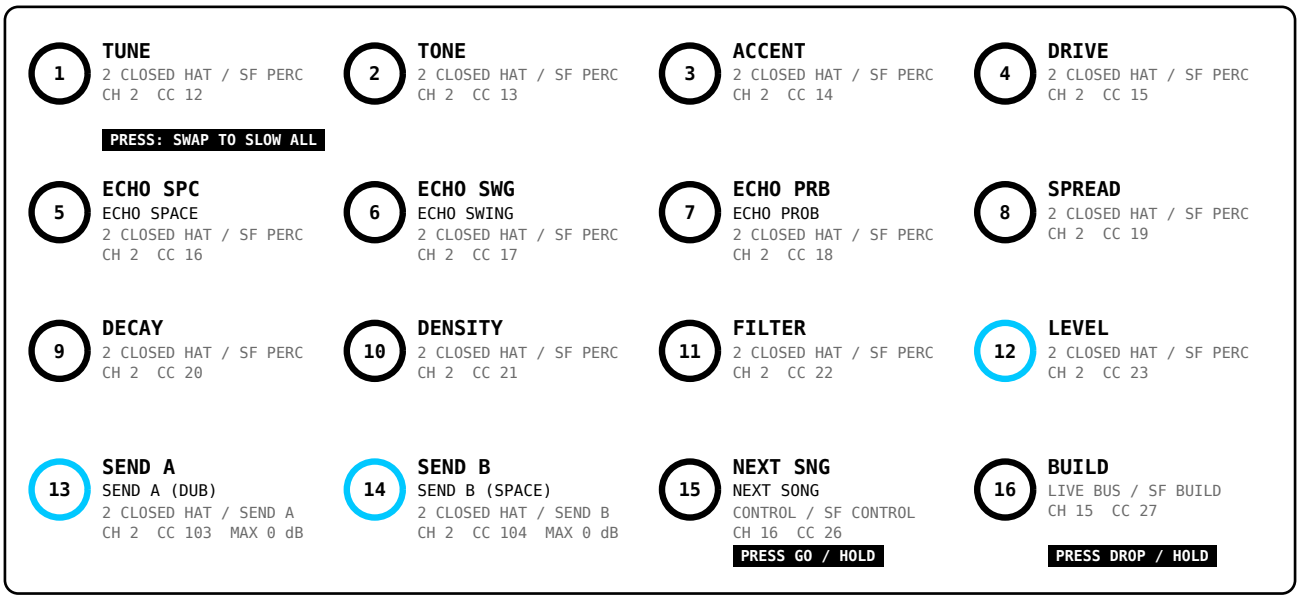
KNOB	SCREEN	WHAT IT DOES
1	TUNE	Kick pitch in semitones around the model's fundamental (about 44-55 Hz).
2	PUNCH	Depth of the pitch sweep. More = harder transient, less = rounder.
3	CLICK	Level of the noise/impulse click on the attack.
4	DRIVE	Saturation on the kick, added to the model's own.
5	PITCH T	How fast the pitch sweep falls. Short = snappy, long = laser.
6	LAYER	Level of the LAYER MODEL. Main and layer always add up to the level of one kick.
7	ECHO SPC	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
8	ECHO SWG	Pushes the first ghost note late by up to a third of ECHO SPACE, so fills sit looser.
9	DECAY	Body length, x0.25 to x4 of the model's decay. This is the kick's GATE. On SAMPLE it gates the sample: 0 = 10 ms, 90 % and up = all of it from START to END.
10	DENSITY	Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
11	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
12	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.
13	SEND A	1 KICK's send into A DUB, the dub delay. Post-fader, up to 0 dB.
14	SEND B	1 KICK's send into B SPACE, the reverb. Post-fader, up to 0 dB.
15	NEXT SNG	Picks the next song; nothing changes until you press it (GO): the song lands on the next bar line. The same knob on every PERF page.
16	BUILD	SF BUILD on LIVE BUS: turn it up over a build. Press it: DROP, on the next bar line (chapter 7). The same knob on every PERF page.

— Set TUNE and DECAY per song and store them: KICK and RUMBLE tune themselves to the song's KEY when KEY is on.

— FILTER right thins the kick, and Pocket closes the low pockets as it thins: the bass and rumble take the sub. Bring the kick's low end back on the drop.

PERF 2 CHH

PERF: SF PERC ON 2 CLOSED HAT, CHANNEL 2



RING: BLACK = A PARAMETER, LIGHT BLUE = A LEVEL. CH / CC: THE ADDRESS.

SF PERC (HAT 909-TYPE) on 2 CLOSED HAT. The fill page.

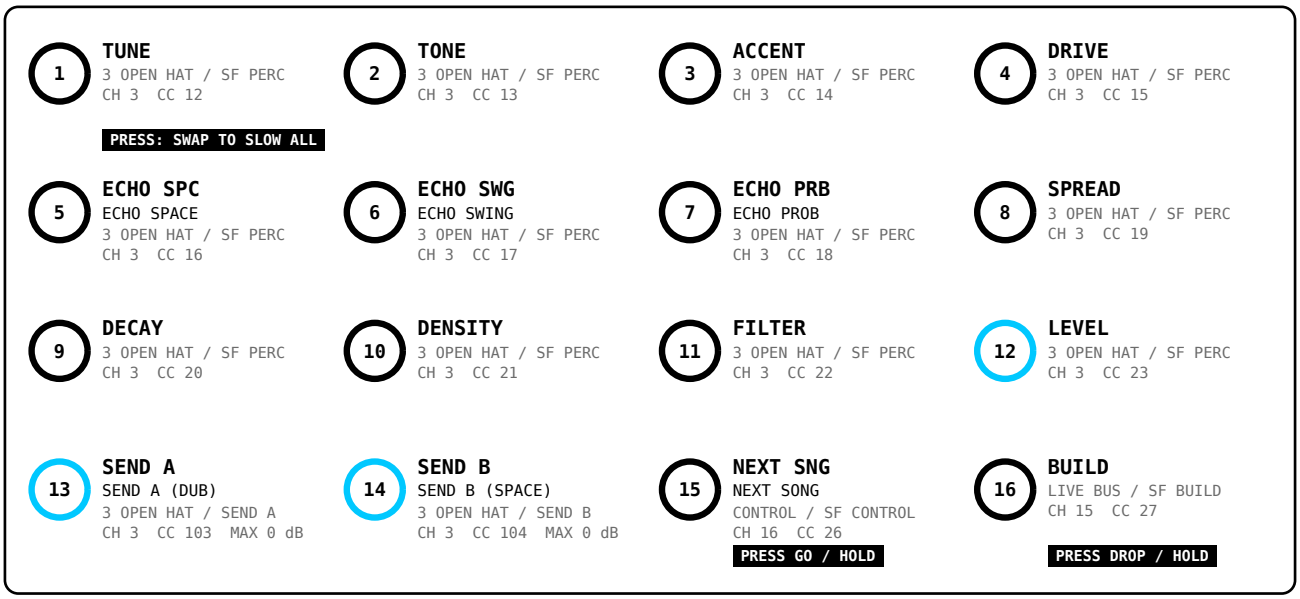
KNOB	SCREEN	WHAT IT DOES
1	TUNE	Pitch in semitones (-24 to +24).
2	TONE	Model-specific brightness: band-pass position on hats, oscillator spread on FM, resonator ratios on MODAL.
3	ACCENT	How much velocity matters. 0 = all hits equal.
4	DRIVE	Saturation on each hit.
5	ECHO SPC	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
6	ECHO SWG	Pushes the first ghost note late by up to a third of ECHO SPACE, so fills sit looser.
7	ECHO PRB	Chance each ghost note plays. Below 100 % the fills become irregular, like a drummer.
8	SPREAD	Pans the DENSITY ghosts alternately left and right.
9	DECAY	Length of each hit, x0.25 to x4. Closed to open. On SAMPLE it gates the sample: 0 = 10 ms, 90 % and up = all of it from START to END.
10	DENSITY	Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
11	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
12	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.
13	SEND A	2 CLOSED HAT's send into A DUB, the dub delay. Post-fader, up to 0 dB.
14	SEND B	2 CLOSED HAT's send into B SPACE, the reverb. Post-fader, up to 0 dB.
15	NEXT SNG	Picks the next song; nothing changes until you press it (GO): the song lands on the next bar line. The same knob on every PERF page.
16	BUILD	SF BUILD on LIVE BUS: turn it up over a build. Press it: DROP, on the next bar line (chapter 7). The same knob on every PERF page.

— DENSITY with ECHO SPACE at 1/16 turns 1/8 hats into 1/16 hats: the send-delay trick, built in. Past 50 % it rolls in 1/32s.

— ECHO SWING and ECHO PROB make the ghosts loose and sparse: a groove that breathes without a new pattern.

PERF 3 OHH

PERF: SF PERC ON 3 OPEN HAT, CHANNEL 3



RING: BLACK = A PARAMETER, LIGHT BLUE = A LEVEL. CH / CC: THE ADDRESS.

SF PERC (HAT 909-TYPE, DECAY 0.8) on 3 OPEN HAT.

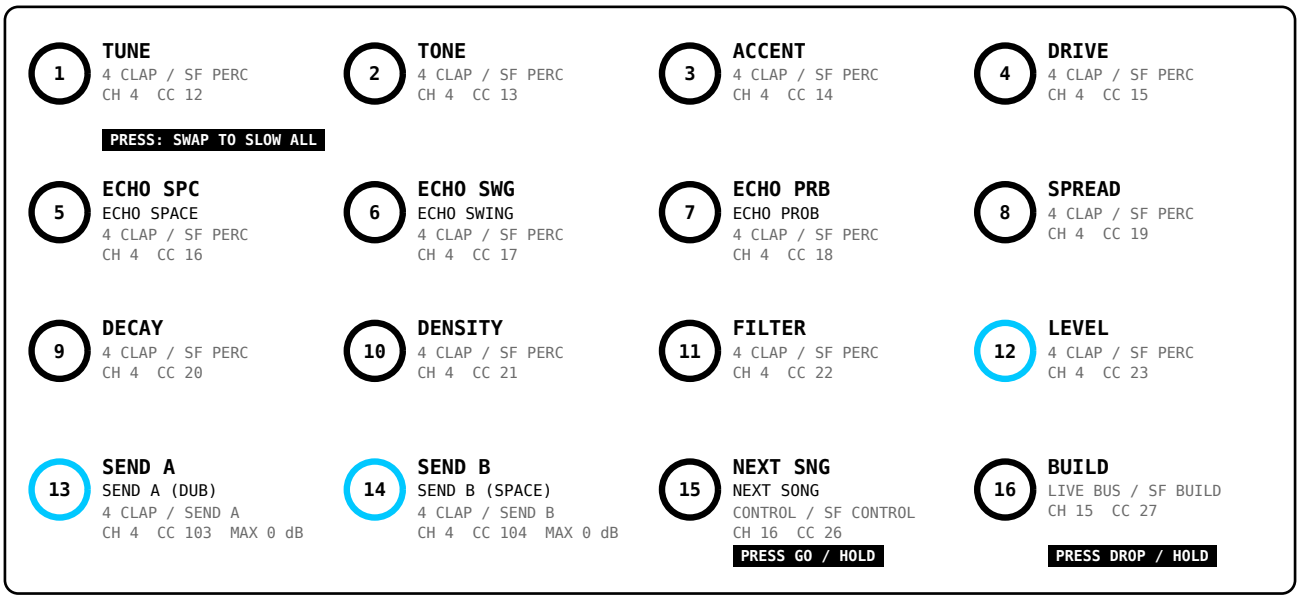
KNOB	SCREEN	WHAT IT DOES
1	TUNE	Pitch in semitones (-24 to +24).
2	TONE	Model-specific brightness: band-pass position on hats, oscillator spread on FM, resonator ratios on MODAL.
3	ACCENT	How much velocity matters. 0 = all hits equal.
4	DRIVE	Saturation on each hit.
5	ECHO SPC	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
6	ECHO SWG	Pushes the first ghost note late by up to a third of ECHO SPACE, so fills sit looser.
7	ECHO PRB	Chance each ghost note plays. Below 100 % the fills become irregular, like a drummer.
8	SPREAD	Pans the DENSITY ghosts alternately left and right.
9	DECAY	Length of each hit, x0.25 to x4. Closed to open. On SAMPLE it gates the sample: 0 = 10 ms, 90 % and up = all of it from START to END.
10	DENSITY	Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
11	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
12	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.
13	SEND A	3 OPEN HAT's send into A DUB, the dub delay. Post-fader, up to 0 dB.
14	SEND B	3 OPEN HAT's send into B SPACE, the reverb. Post-fader, up to 0 dB.
15	NEXT SNG	Picks the next song; nothing changes until you press it (GO): the song lands on the next bar line. The same knob on every PERF page.
16	BUILD	SF BUILD on LIVE BUS: turn it up over a build. Press it: DROP, on the next bar line (chapter 7). The same knob on every PERF page.

— DECAY from closed to open across a build. CHOKE works inside one track only: the closed hat never cuts this one.

— TONE and ACCENT set how the off-beat speaks; SPREAD widens it.

PERF 4 CLP

PERF: SF PERC ON 4 CLAP, CHANNEL 4



RING: BLACK = A PARAMETER, LIGHT BLUE = A LEVEL. CH / CC: THE ADDRESS.

SF PERC (CLAP) on 4 CLAP.

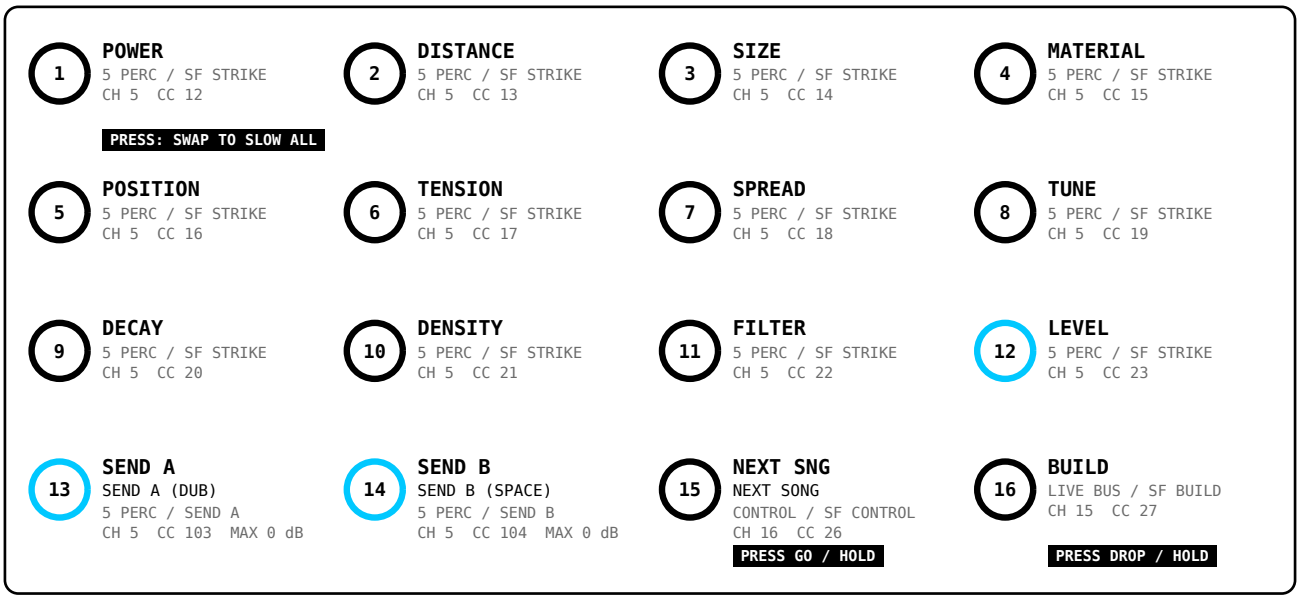
KNOB	SCREEN	WHAT IT DOES
1	TUNE	Pitch in semitones (-24 to +24).
2	TONE	Model-specific brightness: band-pass position on hats, oscillator spread on FM, resonator ratios on MODAL.
3	ACCENT	How much velocity matters. 0 = all hits equal.
4	DRIVE	Saturation on each hit.
5	ECHO SPC	Spacing of the DENSITY ghost notes after each note: 1/8, 1/16, 1/16T, 1/32, 1/8T, 1/8D. 1/8 lands on the offbeat; 1/8D gives a dub gallop.
6	ECHO SWG	Pushes the first ghost note late by up to a third of ECHO SPACE, so fills sit looser.
7	ECHO PRB	Chance each ghost note plays. Below 100 % the fills become irregular, like a drummer.
8	SPREAD	Pans the DENSITY ghosts alternately left and right.
9	DECAY	Length of each hit, x0.25 to x4. Closed to open. On SAMPLE it gates the sample: 0 = 10 ms, 90 % and up = all of it from START to END.
10	DENSITY	Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
11	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
12	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.
13	SEND A	4 CLAP's send into A DUB, the dub delay. Post-fader, up to 0 dB.
14	SEND B	4 CLAP's send into B SPACE, the reverb. Post-fader, up to 0 dB.
15	NEXT SNG	Picks the next song; nothing changes until you press it (GO): the song lands on the next bar line. The same knob on every PERF page.
16	BUILD	SF BUILD on LIVE BUS: turn it up over a build. Press it: DROP, on the next bar line (chapter 7). The same knob on every PERF page.

— DENSITY for rolls into the drop; DECAY short for a snap, long for a room.

— SEND B on the clap's last hit before a breakdown: the room carries it.

PERF 5 PRC

PERF: SF STRIKE ON 5 PERC, CHANNEL 5



RING: BLACK = A PARAMETER, LIGHT BLUE = A LEVEL. CH / CC: THE ADDRESS.

SF STRIKE on 5 PERC: blast-excited bells, plates, bars, tubes and gongs.

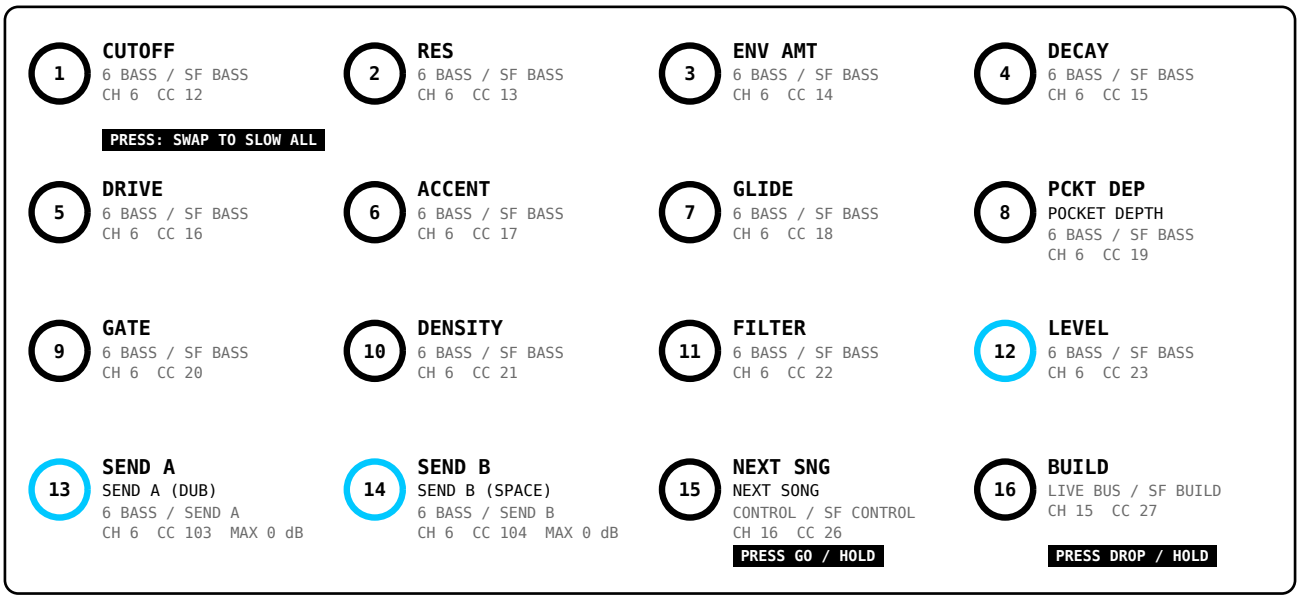
KNOB	SCREEN	WHAT IT DOES
1	POWER	Size of the charge. Velocity scales it (VEL SENS). Above about 70 % the object fractures.
2	DISTANCE	Stand-off. Further = quieter, longer, darker blast that excites fewer high modes.
3	SIZE	Object size, which sets the pitch: the fundamental goes from about 1.4 kHz (small) to 45 Hz (large).
4	MATERIAL	Wood (0) to glass (100 %): how long the high modes ring compared with the low ones.
5	POSITION	Strike point. Striking near a node mutes the modes that have one there, so it changes the colour.
6	TENSION	How sharp a hard hit sounds before it settles (the pitch glides down after the blast).
7	SPREAD	Spreads the modes across the stereo field.
8	TUNE	Pitch in semitones (-24 to +24).
9	DECAY	Ring time, 0.15 to 9 s, times the object's own (a gong rings longer than a bar). Four strikes ring at once; a fifth fades the quietest out over 5 ms.
10	DENSITY	Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
11	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
12	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.
13	SEND A	5 PERC's send into A DUB, the dub delay. Post-fader, up to 0 dB.
14	SEND B	5 PERC's send into B SPACE, the reverb. Post-fader, up to 0 dB.
15	NEXT SNG	Picks the next song; nothing changes until you press it (GO): the song lands on the next bar line. The same knob on every PERF page.
16	BUILD	SF BUILD on LIVE BUS: turn it up over a build. Press it: DROP, on the next bar line (chapter 7). The same knob on every PERF page.

— SIZE sets the pitch; MATERIAL goes from wood to glass; DISTANCE darkens and lengthens the hit.

— POWER above about 70 % fractures the object: save it for the peak.

PERF 6 BAS

PERF: SF BASS ON 6 BASS, CHANNEL 6



RING: BLACK = A PARAMETER, LIGHT BLUE = A LEVEL. CH / CC: THE ADDRESS.

SF BASS on 6 BASS. CUTOFF, RES and ENV AMT move the line; GATE shortens it.

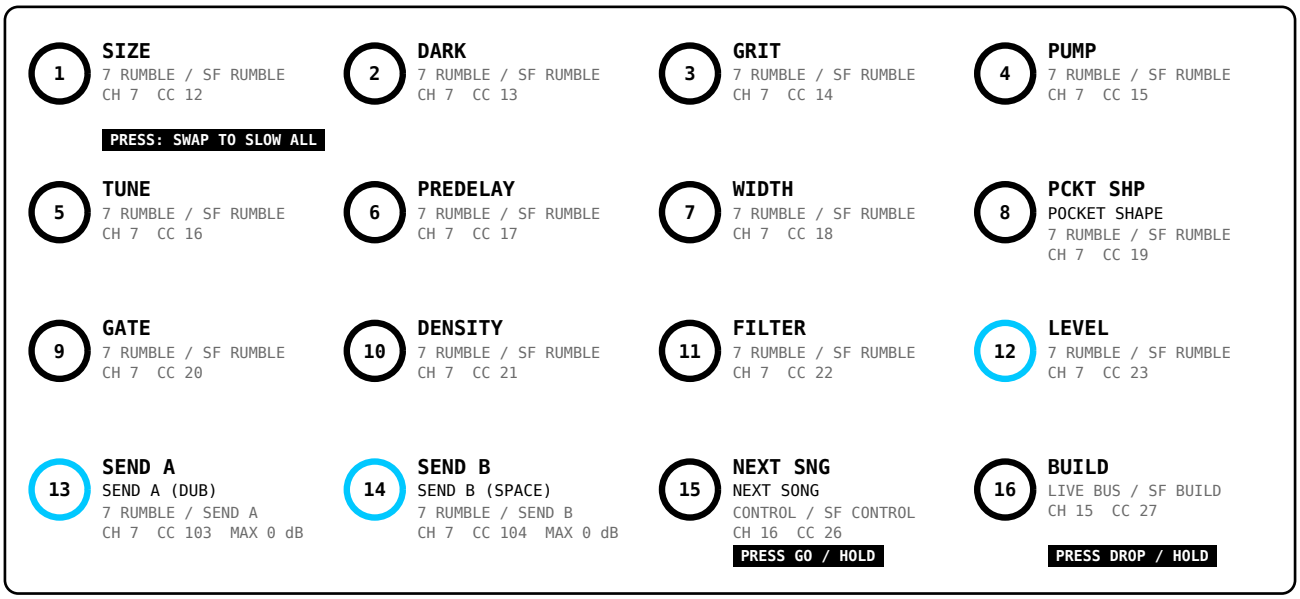
KNOB	SCREEN	WHAT IT DOES
1	CUTOFF	Cutoff of the model's own filter (ladder, state-variable or MS-20-style). Responds to key position and the filter envelope.
2	RES	Resonance of the model's filter. On ACID 303-TYPE this is the squelch.
3	ENV AMT	How far the filter envelope opens the filter on each note. Accented notes open further.
4	DECAY	Length of the filter envelope (and amp decay on most models). Short = plucky, long = sweeping. ACID: the classic decay knob.
5	DRIVE	Saturation before the filter. Adds weight at low settings, grit at high.
6	ACCENT	How strongly accented notes (velocity above about 108) stand out: louder, brighter, shorter filter decay.
7	GLIDE	Portamento between notes (non-acid models). ACID slides only when notes overlap, like a 303.
8	PCKT DEP	How deep this module ducks around SF KICK (Pocket). 0 = no ducking.
9	GATE	Note length. 0 = clipped staccato, 50 % = as played, 100 % = held long (release up to 8x).
10	DENSITY	Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
11	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
12	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.
13	SEND A	6 BASS's send into A DUB, the dub delay. Post-fader, up to 0 dB.
14	SEND B	6 BASS's send into B SPACE, the reverb. Post-fader, up to 0 dB.
15	NEXT SNG	Picks the next song; nothing changes until you press it (GO): the song lands on the next bar line. The same knob on every PERF page.
16	BUILD	SF BUILD on LIVE BUS: turn it up over a build. Press it: DROP, on the next bar line (chapter 7). The same knob on every PERF page.

— GATE down with DENSITY up is a tight, nervous build; GATE open with DENSITY down is spacious dub.

— DRIVE adds weight at low settings and grit at high ones. GLIDE and ACCENT shape the line's slides and pushes.

PERF 7 RMB

PERF: SF RUMBLE ON 7 RUMBLE, CHANNEL 7



RING: BLACK = A PARAMETER, LIGHT BLUE = A LEVEL. CH / CC: THE ADDRESS.

SF RUMBLE on 7 RUMBLE. It plays no notes: it follows SF KICK.

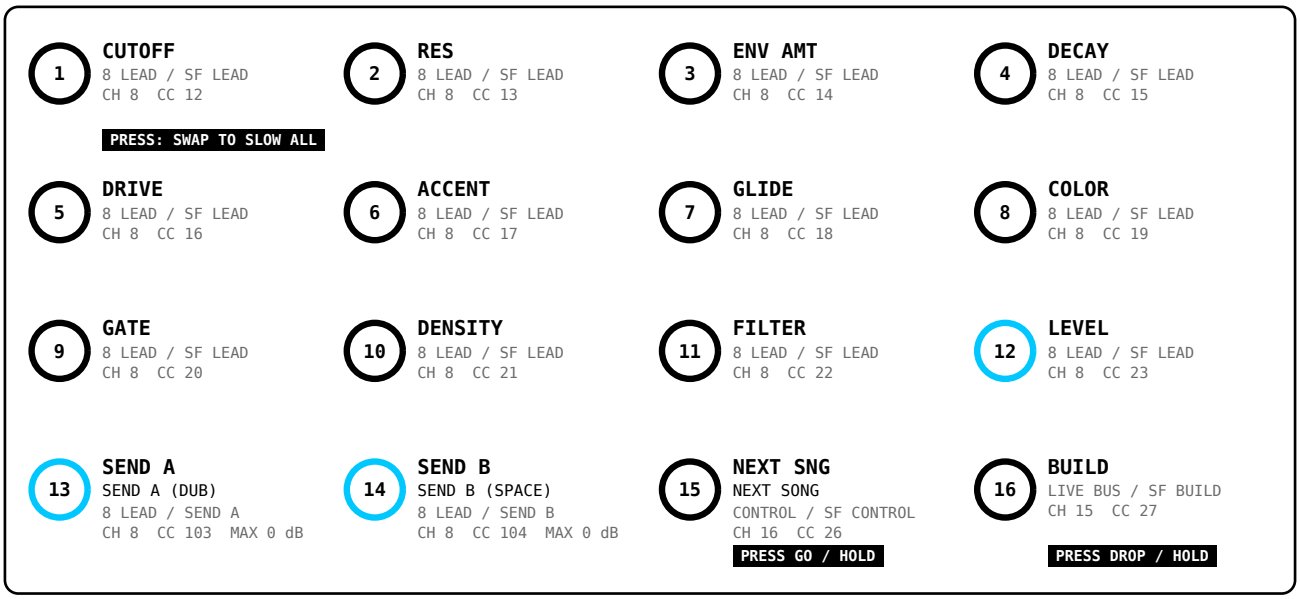
KNOB	SCREEN	WHAT IT DOES
1	SIZE	Reverb size / delay feedback.
2	DARK	Low-pass on the bed: 600 Hz down to 70 Hz.
3	GRIT	Distortion on the bed.
4	PUMP	Pocket depth: how far the bed ducks under each kick.
5	TUNE	Pitch of the thump relative to the kick.
6	PREDELAY	Delay before the reverb: 0, 1/32, 1/16, 1/8.
7	WIDTH	Stereo width of the bed.
8	PCKT SHP	Pocket SHAPE (see Pocket chapter).
9	GATE	Cuts the tail after each kick. 100 % = no gate; lower = shorter.
10	DENSITY	Extra thumps between kicks (offbeat, then 16ths).
11	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
12	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.
13	SEND A	7 RUMBLE's send into A DUB, the dub delay. Post-fader, up to 0 dB.
14	SEND B	7 RUMBLE's send into B SPACE, the reverb. Post-fader, up to 0 dB.
15	NEXT SNG	Picks the next song; nothing changes until you press it (GO): the song lands on the next bar line. The same knob on every PERF page.
16	BUILD	SF BUILD on LIVE BUS: turn it up over a build. Press it: DROP, on the next bar line (chapter 7). The same knob on every PERF page.

— SIZE is the bed's reverb, DARK its low-pass (600 Hz down to 70 Hz), GRIT its distortion; PUMP is how far it ducks under each kick.

— In each section decide whether BASS or RUMBLE carries the sustained low end, then steer that one.

PERF 8 LED

PERF: SF LEAD ON 8 LEAD, CHANNEL 8



RING: BLACK = A PARAMETER, LIGHT BLUE = A LEVEL. CH / CC: THE ADDRESS.

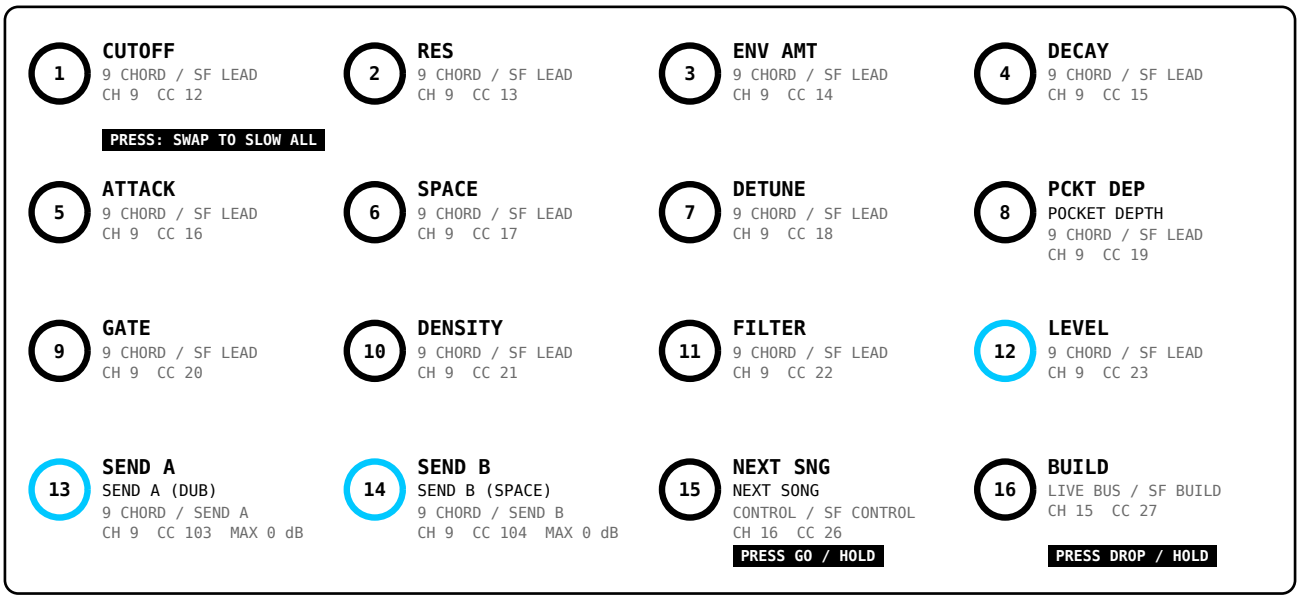
SF LEAD (ACID 303-TYPE) on 8 LEAD. CUTOFF, RES and ENV AMT are the squelch.

KNOB	SCREEN	WHAT IT DOES
1	CUTOFF	Cutoff of the model's own filter (ladder, state-variable or MS-20-style). Responds to key position and the filter envelope.
2	RES	Resonance of the model's filter. On ACID 303-TYPE this is the squelch.
3	ENV AMT	How far the filter envelope opens the filter on each note. Accented notes open further.
4	DECAY	Length of the filter envelope (and amp decay on most models). Short = plucky, long = sweeping. ACID: the classic decay knob.
5	DRIVE	Saturation before the filter. Adds weight at low settings, grit at high.
6	ACCENT	How strongly accented notes (velocity above about 108) stand out: louder, brighter, shorter filter decay.
7	GLIDE	Portamento between notes (non-acid models). ACID slides only when notes overlap, like a 303.
8	COLOR	Model-specific tone control. Mono subtractive: sub-oscillator level. ACID: below 50 % saw, above 50 % square. ORGAN: drawbar brightness. FM: modulation index. SYNC: sync sweep. WAVETABLE: table position (sine > saw > square > vocal). GRANULAR: grain size. NOISE BED: resonance of the noise.
9	GATE	Note length. 0 = clipped staccato, 50 % = as played, 100 % = held long (release up to 8x).
10	DENSITY	Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
11	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
12	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.
13	SEND A	8 LEAD's send into A DUB, the dub delay. Post-fader, up to 0 dB.
14	SEND B	8 LEAD's send into B SPACE, the reverb. Post-fader, up to 0 dB.
15	NEXT SNG	Picks the next song; nothing changes until you press it (GO): the song lands on the next bar line. The same knob on every PERF page.
16	BUILD	SF BUILD on LIVE BUS: turn it up over a build. Press it: DROP, on the next bar line (chapter 7). The same knob on every PERF page.

— Overlapping notes slide like a 303 (always 60 ms): write legato notes on the ONE for slides.
— SLOW ALL at 8 BAR and a quarter turn of CUTOFF: an eight-bar rise that plays itself.

PERF 9 CRD

PERF: SF LEAD ON 9 CHORD, CHANNEL 9



RING: BLACK = A PARAMETER, LIGHT BLUE = A LEVEL. CH / CC: THE ADDRESS.

SF LEAD (STAB DUB) on 9 CHORD. SPACE is the stab's ensemble.

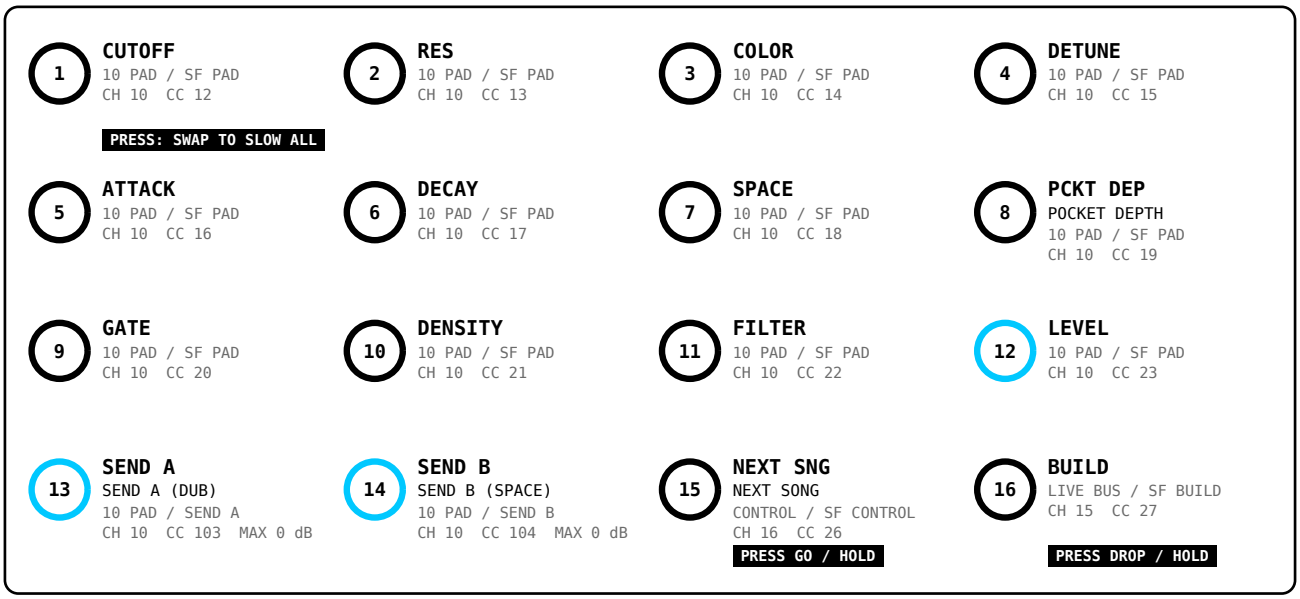
KNOB	SCREEN	WHAT IT DOES
1	CUTOFF	Cutoff of the model's own filter (ladder, state-variable or MS-20-style). Responds to key position and the filter envelope.
2	RES	Resonance of the model's filter. On ACID 303-TYPE this is the squelch.
3	ENV AMT	How far the filter envelope opens the filter on each note. Accented notes open further.
4	DECAY	Length of the filter envelope (and amp decay on most models). Short = plucky, long = sweeping. ACID: the classic decay knob.
5	ATTACK	Adds attack time to amp and filter envelopes. Up for swells.
6	SPACE	Ensemble chorus amount, for the models that have one (strings, poly pad, stabs, hoover, organ).
7	DETUNE	Spread between oscillators and unison voices. Also sets grain scatter on GRANULAR.
8	PCKT DEP	How deep this module ducks around SF KICK (Pocket). 0 = no ducking.
9	GATE	Note length. 0 = clipped staccato, 50 % = as played, 100 % = held long (release up to 8x).
10	DENSITY	Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
11	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
12	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.
13	SEND A	9 CHORD's send into A DUB, the dub delay. Post-fader, up to 0 dB.
14	SEND B	9 CHORD's send into B SPACE, the reverb. Post-fader, up to 0 dB.
15	NEXT SNG	Picks the next song; nothing changes until you press it (GO): the song lands on the next bar line. The same knob on every PERF page.
16	BUILD	SF BUILD on LIVE BUS: turn it up over a build. Press it: DROP, on the next bar line (chapter 7). The same knob on every PERF page.

— If the ONE sends whole chords, set CHORD to NONE on the plugin; if it sends single notes, CHORD voices them. Not both.

— Short GATE plus SEND A (13): the dub-chord move. Ride the DUB return on MIX's LEVELS page.

PERF 10 PAD

PERF: SF PAD ON 10 PAD, CHANNEL 10



RING: BLACK = A PARAMETER, LIGHT BLUE = A LEVEL. CH / CC: THE ADDRESS.

SF PAD (GRANULAR) on 10 PAD. On GRANULAR, COLOR is grain size and DETUNE is scatter.

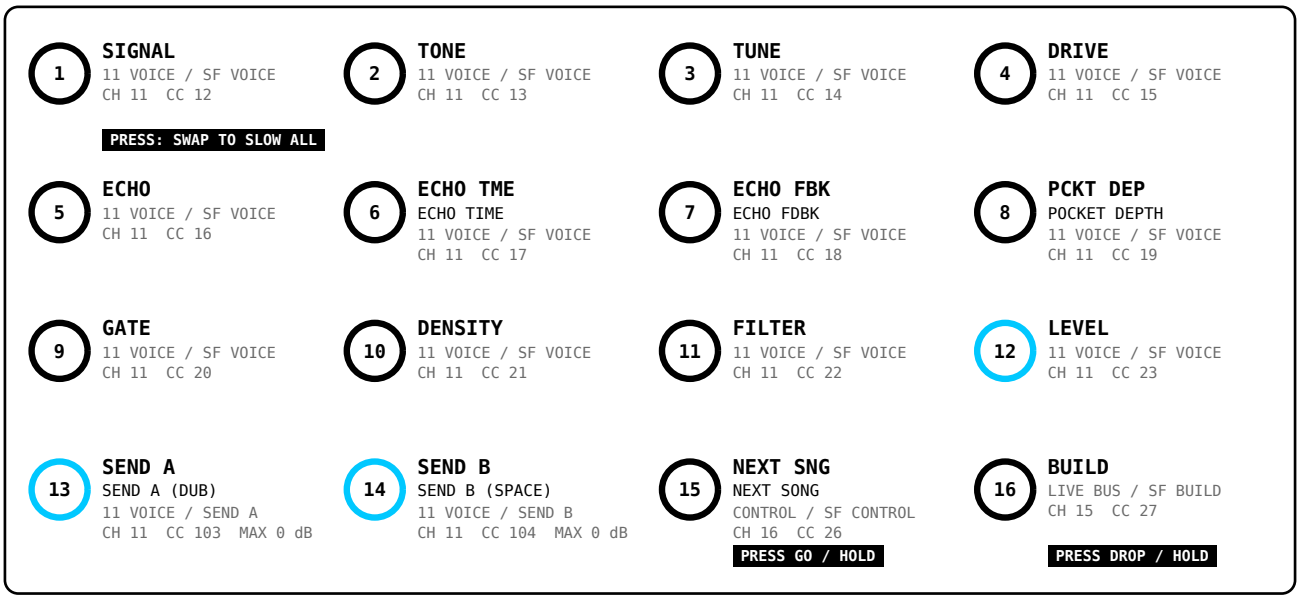
KNOB	SCREEN	WHAT IT DOES
1	CUTOFF	Cutoff of the model's own filter (ladder, state-variable or MS-20-style). Responds to key position and the filter envelope.
2	RES	Resonance of the model's filter. On ACID 303-TYPE this is the squelch.
3	COLOR	Model-specific tone; on GRANULAR, the template's PAD model, it is the grain size.
4	DETUNE	Spread between oscillators and unison voices. Also sets grain scatter on GRANULAR.
5	ATTACK	Adds attack time to amp and filter envelopes. Up for swells.
6	DECAY	Length of the filter envelope (and amp decay on most models). Short = plucky, long = sweeping. ACID: the classic decay knob.
7	SPACE	Ensemble chorus amount, for the models that have one (strings, poly pad, stabs, hoover, organ).
8	PCKT DEP	How deep this module ducks around SF KICK (Pocket). 0 = no ducking.
9	GATE	Note length. 0 = clipped staccato, 50 % = as played, 100 % = held long (release up to 8x).
10	DENSITY	Ghost notes: an echo of every note one ECHO SPACE later, up to 50 %; above it, two more at half that spacing (1/32 rolls). Each ghost quieter, darker, shorter.
11	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
12	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.
13	SEND A	10 PAD's send into A DUB, the dub delay. Post-fader, up to 0 dB.
14	SEND B	10 PAD's send into B SPACE, the reverb. Post-fader, up to 0 dB.
15	NEXT SNG	Picks the next song; nothing changes until you press it (GO): the song lands on the next bar line. The same knob on every PERF page.
16	BUILD	SF BUILD on LIVE BUS: turn it up over a build. Press it: DROP, on the next bar line (chapter 7). The same knob on every PERF page.

— ATTACK up for swells; SPACE is the pad's own room.

— Field recordings work well in GRANULAR: drop one on the plugin before the show.

PERF 11 VOX

PERF: SF VOICE ON 11 VOICE, CHANNEL 11



RING: BLACK = A PARAMETER, LIGHT BLUE = A LEVEL. CH / CC: THE ADDRESS.

SF VOICE on 11 VOICE: phrases, radio fragments, choir.

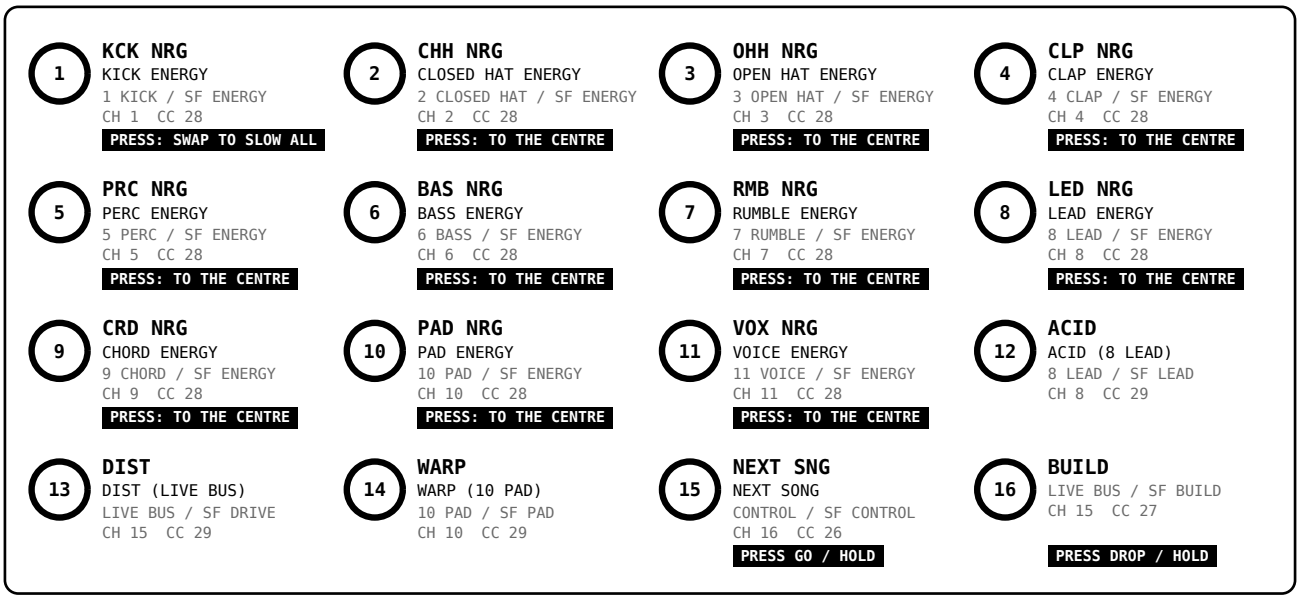
KNOB	SCREEN	WHAT IT DOES
1	SIGNAL	Clarity. RADIO: from lost to clear. Vocoders: sibilance and high-band brightness.
2	TONE	Model-specific: carrier pitch C1-C4 (vocoder, robot, choir), loop length (STUTTER), vowel (choir synth).
3	TUNE	Playback pitch (varispeed) in semitones.
4	DRIVE	Saturation on the voice.
5	ECHO	Send into the built-in dub echo.
6	ECHO TME	Echo time, synced (1 BAR to 1/16). Default 1/4 dotted.
7	ECHO FBK	Echo feedback.
8	PCKT DEP	How deep this module ducks around SF KICK (Pocket). 0 = no ducking.
9	GATE	Slice length. Low values clip each slice short.
10	DENSITY	Re-triggers the slice as ghost repeats (stutter fills).
11	FILTER	Bipolar DJ filter on the output: centre open, left a low-pass down to 60 Hz, right a high-pass up to 12 kHz. A full sweep never jumps in volume.
12	LEVEL	The plugin's output level, stored per song. Set at soundcheck so the volumes at 0 dB give the intended mix.
13	SEND A	11 VOICE's send into A DUB, the dub delay. Post-fader, up to 0 dB.
14	SEND B	11 VOICE's send into B SPACE, the reverb. Post-fader, up to 0 dB.
15	NEXT SNG	Picks the next song; nothing changes until you press it (GO): the song lands on the next bar line. The same knob on every PERF page.
16	BUILD	SF BUILD on LIVE BUS: turn it up over a build. Press it: DROP, on the next bar line (chapter 7). The same knob on every PERF page.

— SIGNAL goes from lost to clear (RADIO model); ECHO feeds the voice's own dub echo, ECHO TME and ECHO FBK shape it.

— C3 plays slice 1, C#3 slice 2 and so on; below C3 the whole phrase. Sequence it as choreography.

PERF 12 NRG

PERF: EVERY SF ENERGY, NEXT SONG AND BUILD



RING: BLACK = A PARAMETER, LIGHT BLUE = A LEVEL. CH / CC: THE ADDRESS.

Every ENERGY knob: knob n is track n's SF ENERGY. Then ACID, DIST and WARP. The arrangement under one hand.

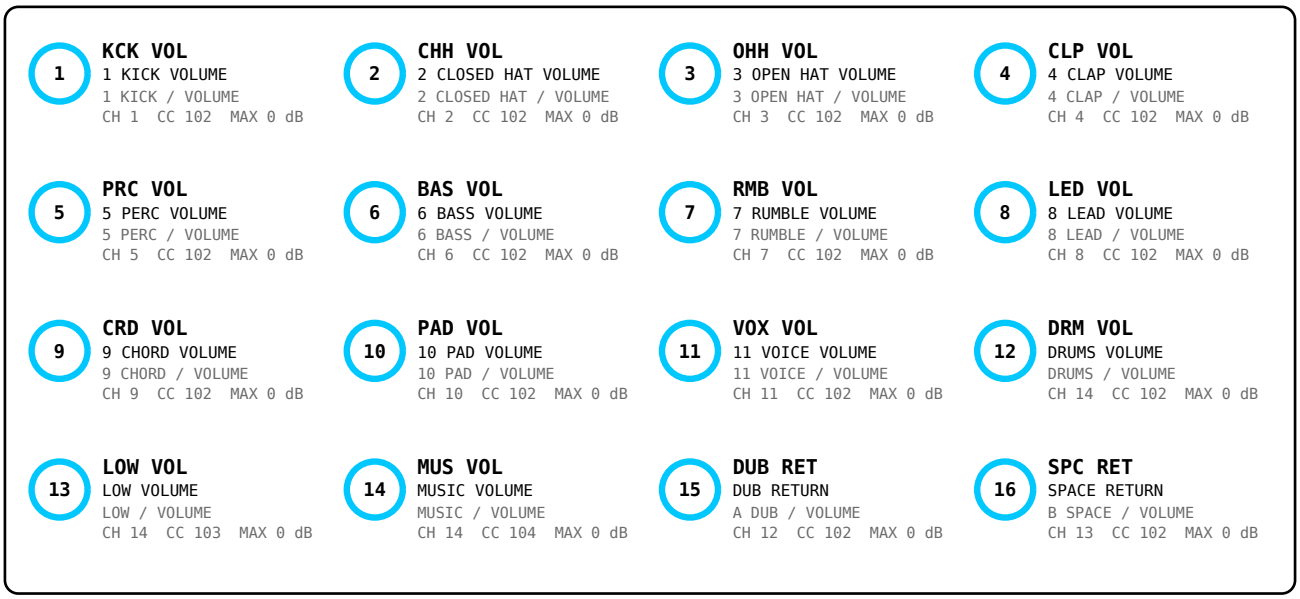
KNOB	SCREEN	WHAT IT DOES
1	KCK NRG	1 KICK's SF ENERGY: centre neutral, left takes energy out, right puts it in.
2	CHH NRG	2 CLOSED HAT's SF ENERGY: centre neutral, left takes energy out, right puts it in.
3	OHH NRG	3 OPEN HAT's SF ENERGY: centre neutral, left takes energy out, right puts it in.
4	CLP NRG	4 CLAP's SF ENERGY: centre neutral, left takes energy out, right puts it in.
5	PRC NRG	5 PERC's SF ENERGY: centre neutral, left takes energy out, right puts it in.
6	BAS NRG	6 BASS's SF ENERGY: centre neutral, left takes energy out, right puts it in.
7	RMB NRG	7 RUMBLE's SF ENERGY: centre neutral, left takes energy out, right puts it in.
8	LED NRG	8 LEAD's SF ENERGY: centre neutral, left takes energy out, right puts it in.
9	CRD NRG	9 CHORD's SF ENERGY: centre neutral, left takes energy out, right puts it in.
10	PAD NRG	10 PAD's SF ENERGY: centre neutral, left takes energy out, right puts it in.
11	VOX NRG	11 VOICE's SF ENERGY: centre neutral, left takes energy out, right puts it in.
12	ACID	One knob for the squelch (the E16's NRG 12, on 8 LEAD): RES towards 95 %, ENV AMT towards 90 %, ACCENT towards full, DRIVE +35 % and CUTOFF +10 %, each from where its own knob is (never down). Made for ACID 303-TYPE; any model takes it. 0 = off.
13	DIST	Dry/wet. The dry signal runs through an identical oversampler, so the blend is phase-aligned (no comb filtering) and MIX 0 is the clean input.
14	WARP	One knob for a melting texture (the E16's NRG 14, on 10 PAD): tape wow (up to 35 cents, two slow drifts) and flutter (5 cents), every voice on its own phase and pace so a chord smears; DETUNE and SPACE pushed up too (on GRANULAR: the grains scatter wider). 0 = off.
15	NEXT SNG	Picks the next song; nothing changes until you press it (GO): the song lands on the next bar line. The same knob on every PERF page.
16	BUILD	SF BUILD on LIVE BUS: turn it up over a build. Press it: DROP, on the next bar line (chapter 7). The same knob on every PERF page.

— Centre is neutral: left takes energy out (filter, space), right puts it in (drive, echo, chop). Small moves, often.

- Press an ENERGY knob and it goes back to the centre; with SLOW ALL on it glides there.
- KCK NRG all the way down takes the kick out of the mix (the drum tracks' ENERGY fades out over its last eighth).
- 12 ACID: 8 LEAD's 303 squelches (RES, ENV AMT, ACCENT, DRIVE and CUTOFF up together). 13 DIST: the whole mix through SF DRIVE on the LIVE BUS, its lows clean, its level held. 14 WARP: 10 PAD wobbles and smears (tape wow and flutter, spread, ensemble). All three off at zero; their presses do nothing.

MIX 1 LEVELS

MIX: THE VOLUMES, IN NRG'S PLACES



RING: BLACK = A PARAMETER, LIGHT BLUE = A LEVEL. CH / CC: THE ADDRESS.

MIX. The volumes in NRG's places: knob n is track n's, 12-14 the groups', 15 and 16 the returns'.

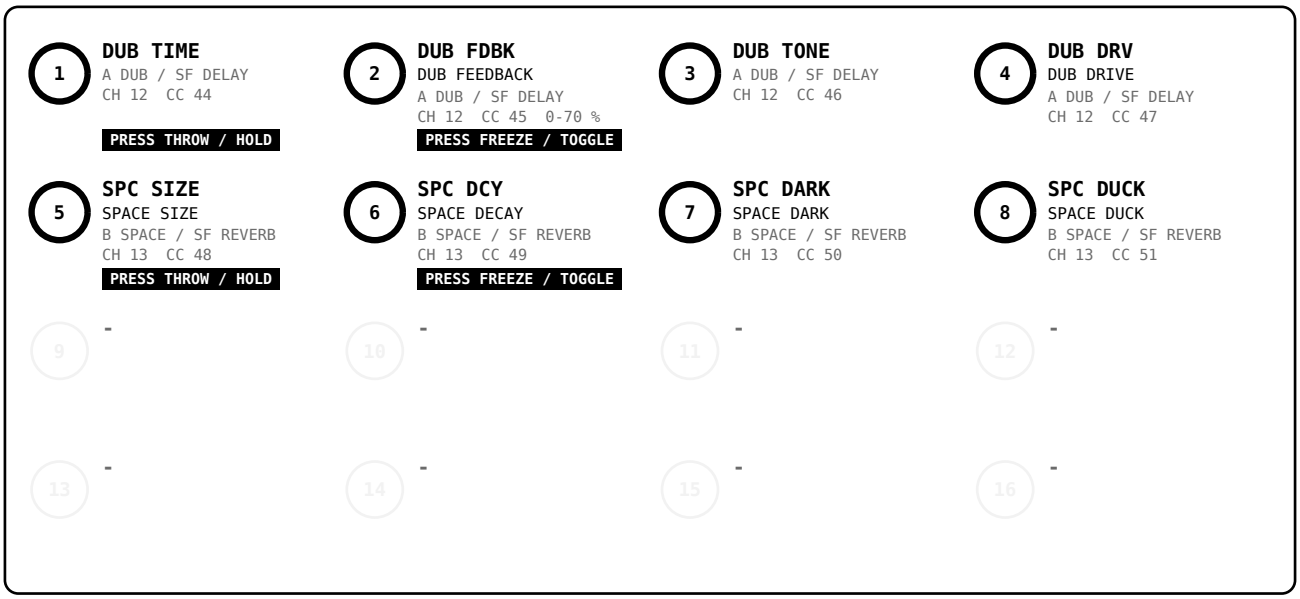
KNOB	SCREEN	WHAT IT DOES
1	KCK VOL	The volume of 1 KICK, up to 0 dB.
2	CHH VOL	The volume of 2 CLOSED HAT, up to 0 dB.
3	OHH VOL	The volume of 3 OPEN HAT, up to 0 dB.
4	CLP VOL	The volume of 4 CLAP, up to 0 dB.
5	PRC VOL	The volume of 5 PERC, up to 0 dB.
6	BAS VOL	The volume of 6 BASS, up to 0 dB.
7	RMB VOL	The volume of 7 RUMBLE, up to 0 dB.
8	LED VOL	The volume of 8 LEAD, up to 0 dB.
9	CRD VOL	The volume of 9 CHORD, up to 0 dB.
10	PAD VOL	The volume of 10 PAD, up to 0 dB.
11	VOX VOL	The volume of 11 VOICE, up to 0 dB.
12	DRM VOL	The DRUMS group's volume: 2-5, the hats, the clap and the perc (not the kick), up to 0 dB.
13	LOW VOL	The LOW group's volume: 6 BASS and 7 RUMBLE, up to 0 dB.
14	MUS VOL	The MUSIC group's volume: 8-11, up to 0 dB.
15	DUB RET	The dub return's volume (SF DELAY on A DUB), up to 0 dB.
16	SPC RET	The reverb return's volume (SF REVERB on B SPACE), up to 0 dB.

— Every volume is mapped up to 0 dB, so a full turn never runs past the soundcheck balance.

— Set each instrument's LEVEL (its song) so the volumes at 0 dB give the intended mix; the volumes are your balance on the night.

MIX 2 SPACE

MIX: SF DELAY ON A DUB, SF REVERB ON B SPACE



RING: BLACK = A PARAMETER, LIGHT BLUE = A LEVEL. CH / CC: THE ADDRESS.

MIX. The returns: 1-4 DUB (SF DELAY on A DUB), 5-8 SPACE (SF REVERB on B SPACE). The presses sit under their knobs.

KNOB	SCREEN	WHAT IT DOES
1	DUB TIME	A DUB: Delay time, synced: 1 BAR to 1/32, then 5/16 and 7/16 (odd lengths: the repeats walk across the beat). Changing it slews the head (pitch bend).
2	DUB FDBK	A DUB: Repeats, up to 115 %. Above 100 % the loop saturates instead of growing.
3	DUB TONE	A DUB: Bipolar filter inside the feedback loop: left darker, right thinner. Each repeat goes further than the last.
4	DUB DRV	A DUB: Saturation inside the feedback loop (2x oversampled): each repeat dirtier than the last.
5	SPC SIZE	B SPACE: Room size.
6	SPC DCY	B SPACE: Decay time of the lows, 0.3 to 30 s (measured: DECAY is the real RT60).
7	SPC DARK	B SPACE: Damping: shortens the decay of the highs only, so the tail gets darker without getting shorter.
8	SPC DUCK	B SPACE: Ducks the tail under the kick and while the dry signal plays.

— THROW (hold) sends into the effect for as long as you hold it. FREEZE (toggle) loops what is in it until pressed again.

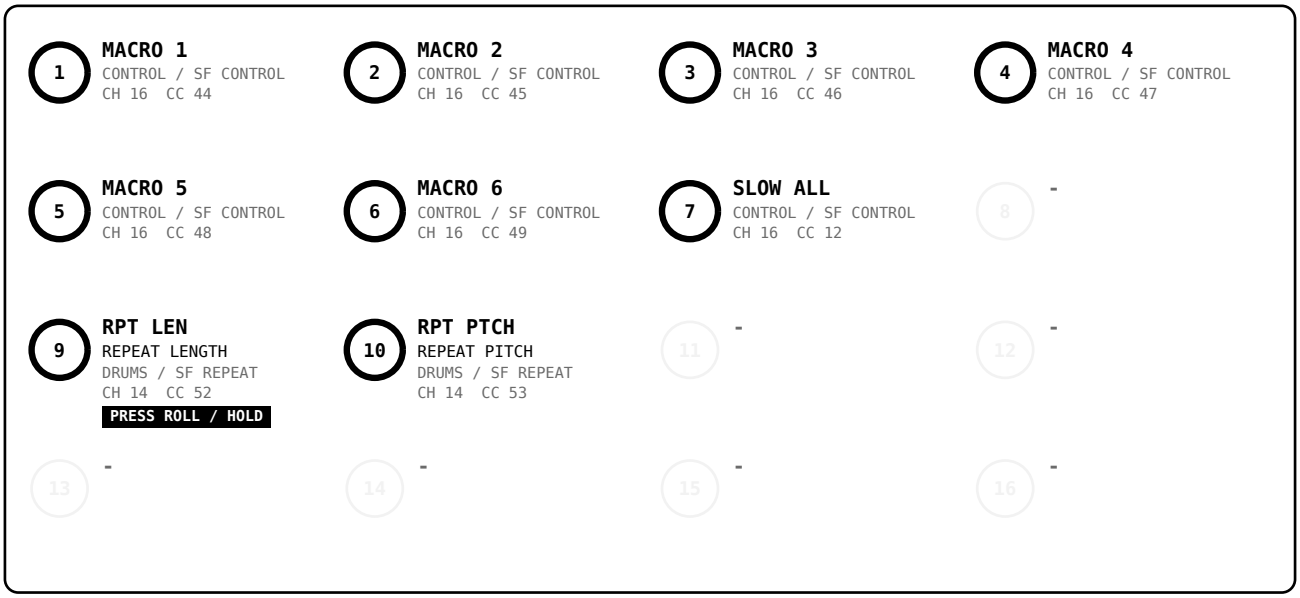
— Turn DUB TIME while frozen: the loop bends like tape speed and locks again when you let go.

— DUB FEEDBACK is mapped 0-70 % at first; widen the mapping when the music asks.

— SF TERROIR (your captured places) can replace SF REVERB on B SPACE: the same four knobs and two presses, mapped once (Cmd-M).

MIX 3 MACROS

MIX: SF CONTROL'S MACROS, SF REPEAT ON DRUMS



RING: BLACK = A PARAMETER, LIGHT BLUE = A LEVEL. CH / CC: THE ADDRESS.

MIX. SF CONTROL's six macros and SLOW ALL, and SF REPEAT on DRUMS.

KNOB	SCREEN	WHAT IT DOES
1	MACRO 1	Macro 1: every SF plugin with MACRO SRC = MACRO 1 follows it. ENERGY in the starting plan: the group ENERGY plugins and BUILD together.
2	MACRO 2	Macro 2: every SF plugin with MACRO SRC = MACRO 2 follows it. MOOD: PAD, CHORD and VOICE (cutoff, envelope, drive; the voice's SIGNAL and ECHO).
3	MACRO 3	Macro 3: every SF plugin with MACRO SRC = MACRO 3 follows it. DENSITY: CLOSED HAT, OPEN HAT, CLAP and PERC (DECAY and DENSITY: the fills).
4	MACRO 4	Macro 4: every SF plugin with MACRO SRC = MACRO 4 follows it. PULSE: KICK and RUMBLE (kick DECAY, PUNCH, DRIVE; rumble SIZE, GRIT, DENSITY).
5	MACRO 5	Macro 5: every SF plugin with MACRO SRC = MACRO 5 follows it. SPACE: the DUB and SPACE returns (delay FEEDBACK and MIX; reverb SIZE, DECAY and MIX).
6	MACRO 6	Macro 6: every SF plugin with MACRO SRC = MACRO 6 follows it. FRACTURE: LEAD and BASS (cutoff, envelope, density and drive: smooth to heavy).
7	SLOW ALL	How long your next knob moves take (OFF, 1-128 BARS) in every SF plugin whose SLOW is CONTROL. A move keeps its length: turning SLOW ALL never touches a travelling knob. A move in the same audio buffer as a SLOW ALL change takes the length from before it, in every plugin alike (with the transport stopped, the new length counts at once). Song recalls follow it. Only while SF CONTROL runs.
9	RPT LEN	Loop length: 1/4 to 1/32 and triplets.
10	RPT PTCH	Pitch drop over the repeats: falls by this amount every four passes.

— Hold ROLL on 9 and turn LENGTH under the same finger: the drums roll at the length you dial.

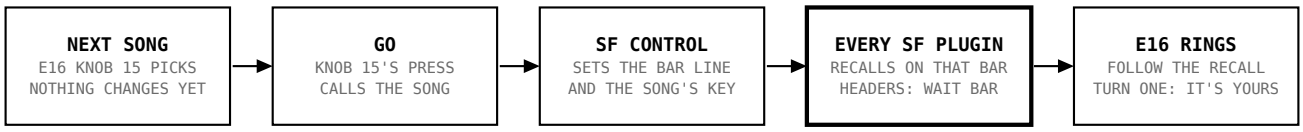
— 10 PITCH makes the roll fall as it repeats. The macro plan is in chapter 7. SLOW ALL is a plain knob here (7).

6 SONGS, PATTERNS AND SCENES

Each system owns one thing. The ONE owns the notes, SF songs own the sound, the E16 owns where your hands are. A song change is one knob and one press: every SF plugin lands on the same bar.

SYSTEM	RESPONSIBLE FOR
OXI ONE projects and patterns	Notes, rhythms, gates, accents, the pattern progression.
SF songs (SF CONTROL)	The sound of every SF plugin and the song's KEY, called from the E16 and recalled on the bar. Sixteen songs, saved with the set.
E16 pages	Where your hands are. Pages hold no sound: changing pages changes nothing you hear.
Live Session scenes	BRIDGE clips, reference material, rehearsal.
Live Arrangement	Recording the performance.

A SONG CHANGE



— **NEXT SONG** (knob 15 of any PERF page) picks the song and does nothing else: a knob knocked in passing never changes the song. SF CONTROL shows NEXT in amber, PRESS GO, until you call it.

— **GO** (its press) calls it: SF CONTROL decides the bar line once and sends it with the song, so every plugin lands on the same bar whatever order Live processes the tracks in. Pressed up to 30 ms after a bar line, it counts for that bar; later, for the next one. With the transport stopped it lands at once. Every plugin's header shows WAIT BAR in green until it lands.

— GO again, or GO on the song already in force, does nothing. NEXT SONG follows any song change (a turn of SONG on SF CONTROL, STORE TO), so a pick you didn't call gives way.

— A song and a ONE project are independent: any song with any project's patterns. A plugin with nothing stored in that song stays as it is.

— **SLOW ALL** on SF CONTROL makes a recall glide: at 8 or 16 BAR every SF knob the song stores travels to it over 8 or 16 bars; at OFF it lands on the bar. Menus and switches (a MODEL, an ECHO SPACE) land on the bar either way (GLIDES, chapter 7, has the list).

SAVING A SONG

— Set every plugin how the song should start, then press **STORE ALL** on SF CONTROL: every SF plugin stores its values into the song in force (SONG).

— **Into another song:** choose it in **STORE TO** first, then STORE ALL. That song isn't recalled, so nothing you hear changes; it becomes the song in force (SONG follows) and STORE TO goes back to ACTIVE SONG. Choosing it on SONG instead would recall it and lose what you set.

— If a song change is under way when you STORE TO (WAIT BAR), the change still lands and stays the song in force.

— STORE in a plugin's header stores just that plugin. Nothing that rewrites a song is on the E16.

A TRACK IN THE SET

1. Turn NEXT SONG (15) to the track's song and press it on the bar before the change: the song lands on the bar.
2. Load the track's ONE project and play its patterns in order (SPARSE, MAIN, VARIATION, EXIT); play the sound from the E16.
3. Before the next track: BUILD and a drop, a dub throw or a BRIDGE clip covers the change (chapter 7).

7 PERFORMANCE

NRG is the arrangement, a track's page is the detail. Very simple note content, all the variation played on knobs: every move below is a knob or a press, on the bar.

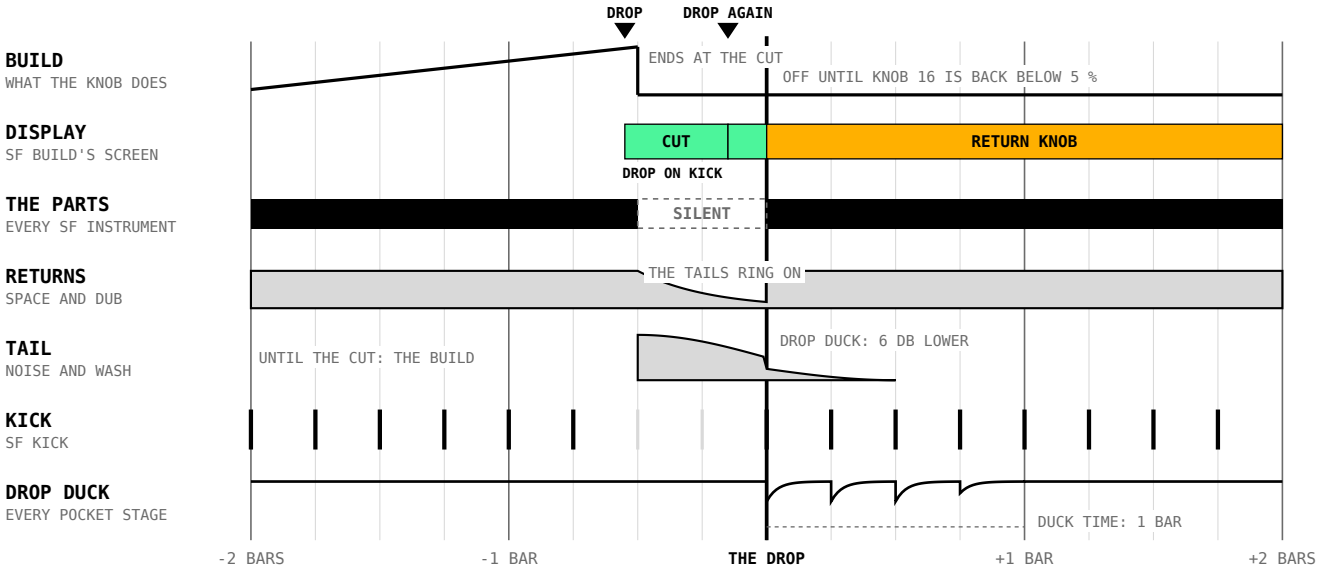
THE SET ARC

PHASE	WHAT CARRIES IT	WHERE
OPENING	Kick and rumble alone; PAD or VOICE far back. ENERGY knobs left of centre.	NRG; RMB, PAD
LOCK-IN	Hats come in by DENSITY, not by level. One part at a time, eight or sixteen bars apart.	CHH, KCK
BUILD	BUILD up over 8-16 bars; hats and clap DENSITY up; SPACE DUCK down so the tail fills the bar.	Knob 16; CHH, OHH, CLP; MIX SPACE
PEAK	The drop; ENERGY knobs right of centre; the lead or the chord carries the hook; dub throws on the offbeats.	Press 16, NRG, LED, SPACE
BREAKDOWN	KCK NRG to zero (the kick out); BAS NRG down; WARP up on the pad; FREEZE a return; VOICE up.	NRG, SPACE, VOX
RETURN	Kick back on the one; a song change lands here (NEXT SONG, then GO).	Press 16, knob 15, the ONE
CLOSE	ENERGY knobs left; parts out one by one; the last bars on a frozen reverb or BRIDGE.	NRG, SPACE, Live's BRIDGE

A starting plan, not a rule: hypnotic sets hold a phase far longer than feels safe.

BUILDS AND DROPS

BUILD is one knob for a whole-mix build (16, on every PERF page): a rising high-pass, a noise riser, a reverb wash and width. The drop is played on its press (DROP): press once and the music stops, leaving the tails; press again and the drop comes in on the next kick. The gap is as long as you play it. The drawing: DROP half a bar before the drop, DROP again just before it; TAIL 1 BAR, DROP DUCK 6 dB, DUCK TIME 1 BAR.



— **DROP**, the first press: from the next sixteenth every SF instrument stops at its source (KICK, PERC, STRIKE, RUMBLE, BASS, LEAD, PAD, VOICE, and STRIP and TAP unless their ROLE is DECK). Their sends go quiet, so SPACE's reverb and DUB's echoes ring out; the build ends there and its noise and wash ring for TAIL. DECK and BRIDGE play on. The parts keep playing silently: the drop comes in mid-pattern, as a mute would. SF KICK reports nothing while silent, so no tail ducks to a kick nobody hears.

— **DROP again**: the drop comes in on the next beat, on the kick. The parts come back 10 ms before it, so the kick keeps its attack; a part with a note up to 30 ms early comes back just before that note. Two quick presses give a sixteenth's cut into the next beat. A third press while the drop waits changes nothing; a press after the drop cuts again.

— Live stopping, a jump back (an Arrangement loop wrapping), PANIC, or SF BUILD switched off end the cut at once (within a second for BUILD switched off).

— **TAIL** (CUT to 2 BARS): the riser's noise and the wash ring on after the build ends at the cut, fading out; longer than the gap, they carry into the drop, DROP DUCK lower.

— **DROP DUCK** (0-12 dB, off by default) and **DUCK TIME** (1/4 to 4 BARS): after the drop every SF part with Pocket dips under each kick, whatever its POCKET DEPTH: bass, lead, pad, voice, rumble, both returns. The first kicks stand alone on top of everything that comes back, with less weight than a kick brought back by hand: the bass and rumble around them are gone.

— **LAND ON** (KICK or BAR LINE): KICK, the default, puts the cut and the drop where SF KICK's kicks land against Live's beat (a sequencer clocked by Live plays late: the OXI One about 40 ms), and BUILD's readout under its bar shows where the kick reaches it (KICK +80 MS). Until the kick is found, and on BAR LINE: Live's beat.

— After a drop the build stays off until BUILD (16) is back below 5 % (display: RETURN KNOB). Bring it down in your own time: nothing re-builds.

RECIPE	THE GAP	TAIL	DROP DUCK	DUCK TIME	WHAT IT DOES
HARD STOP	1 BEAT	CUT	OFF	-	DROP on the beat before, DROP again: a beat of air, then everything on the one.
BREATH	2 BEATS	1/4	OFF	-	The riser rings into the gap and dies before the kick; the reverb hangs.
TAP	1/16	1 BAR	6 dB	1 BAR	Two quick presses just before the one: a sixteenth's cut, the wash under the first bar, ducked.
LONG FALL	1 BAR+	1/2	9 dB	2 BARS	DROP a bar out and wait: the space rings; DROP again when it feels right.
SNAP	-	-	OFF	-	No DROP: turn BUILD down on the bar line; nothing else changes.

Defaults: TAIL 1/8, DROP DUCK OFF, DUCK TIME 1 BAR, LAND ON KICK. UNUSED in Live's list was GAP. The drop duck needs KICK REPORTING on SF CONTROL. TAP and STRIP set to ROLE KICK, PERC or DECK don't duck.

ENERGY AND MACROS

— The ENERGY knobs on NRG are the fastest way to move a whole set: small moves, often, a track's own knob one part. ACID, DIST and WARP (12-14) are the set's three colour moves: the 303 squeals, the mix grits up, the pad melts. The groups' ENERGY plugins still follow MACRO 1 (below); a negative MACRO 1 amount on LOW darkens the bass while the rest lifts.

— Six macros on MIX's MACROS page, each followed by the plugins that subscribe to it (MACRO SRC, MACRO AMT on each plugin). A starting plan:

MACRO	FOLLOWED BY	MOVES
1 ENERGY	the group ENERGY plugins, BUILD	every group's energy at once

MACRO	FOLLOWED BY	MOVES
2 MOOD	PAD, CHORD, VOICE	cutoff, envelope and drive; the voice's SIGNAL and ECHO
3 DENSITY	CLOSED HAT, OPEN HAT, CLAP, PERC	DECAY and DENSITY: the fills
4 PULSE	KICK, RUMBLE	kick DECAY, PUNCH, DRIVE; rumble SIZE, GRIT, DENSITY
5 SPACE	DUB, SPACE returns	delay FEEDBACK and MIX; reverb SIZE, DECAY and MIX
6 FRACTURE	LEAD, BASS	cutoff, envelope, density and drive: smooth to heavy

Start the amounts at 30-50 %. One macro per plugin.

DUB

— **Sends:** every track page has the track's two sends (13 into A DUB, 14 into B SPACE). A quarter turn of 9 CHORD's SEND A on the last stab of a phrase is the dub-chord move.

— **Throws:** MIX's SPACE page, hold 1 (DUB THROW) on an offbeat stab or a voice fragment and let go: the repeats carry on. With INPUT THROW ONLY on SF DELAY, the press is the only way in.

— **Tape moves:** turn DUB TIME while repeats are running: the head slews and the pitch bends. FREEZE (2) then TIME: the frozen loop bends like tape speed.

— **Repeats that grow:** DUB FEEDBACK past 100 % saturates instead of exploding; the mapping stops at 70 % until you widen it.

— **Space that breathes:** SPACE DUCK (8) 30-60 % lets the reverb bloom in the gaps between kicks. FREEZE (6) holds a chord in the room for a breakdown.

FILLS AND TRANSITIONS

— **Hats:** CHH, the closed hat's page: DENSITY (10) up with ECHO SPACE (5) at 1/16: 1/8 hats become 1/16s, then 1/32 rolls past 50 %.

— **Drum roll:** MIX's MACROS page, hold ROLL (9) and turn LENGTH under the same finger: 1/4 down to 1/32. PITCH (10) makes it fall.

— **Between tracks:** turn NEXT SONG (15) during a build or a breakdown and press it on the bar before the change; load the next ONE project when you like, the two are independent. With SLOW ALL set, the knobs glide into the new song; a MODEL change in it still lands on the bar, so cover it with the build or the drop.

— **BRIDGE:** bring BRIDGE up and LIVE BUS down in Live to hand the room to a clip while you regroup. BRIDGE still needs Live: it is not a backup for a computer failure.

GLIDES

— **A travelling knob's ring turns green** and walks with the sound to where you set it, then turns white. Turn it while it walks: the ring shows your hand, then green again. (SF CONTROL's E16 RINGS and the scene's script; nothing else depends on them.)

— **SLOW ALL** (press knob 1 of any PERF page, turn, press again; or MACROS 7 on MIX) sets how long your next moves take, in every SF plugin. OFF: they land at once. 8 BAR: a quarter turn of LEAD CUTOFF becomes an eight-bar rise that plays itself.

— **A move keeps its length.** Start a 16-bar rise on the pad, set SLOW ALL to OFF, and make quick moves elsewhere: the pad keeps rising. Turn a travelling knob and it sets off again from where it has got to.

— **SLOW ALL changed in the same instant as a move** (one audio buffer, a few ms): the move takes the length from before the change, in every plugin alike. Set SLOW ALL, then move.

— **Song changes follow it, knob by knob.** SLOW ALL at 16 BAR before GO: every SF knob the song stores travels to the new song over 16 bars; OFF: it lands on the bar. What does not glide (the table) changes on the bar. It is a morph of the knobs, not a crossfade between two whole sounds.

— A plugin's own SLOW, set in its window, wins over SLOW ALL; leave it at CONTROL.

WHAT SLOW MOVES (every knob and press of PERF and MIX; drawn from the scenes and the plugins)

ON THE E16	WITH SLOW / SLOW ALL
Every SF knob, 155 of them: each plugin's top row, and the details on the scenes too (BASS ACCENT, BASS DECAY, BASS GLIDE, BASS POCKET DEPTH, CHORD ATTACK, CHORD DECAY, CHORD DETUNE, CHORD POCKET DEPTH, CHORD SPACE, CLAP ECHO PROB, CLAP ECHO SWING, CLAP SPREAD, CLOSED HAT ECHO PROB, CLOSED HAT ECHO SWING, CLOSED HAT SPREAD, KICK ECHO SWING, KICK LAYER, KICK PITCH TIME, LEAD ACCENT, LEAD ACID, LEAD COLOR, LEAD DECAY, LEAD GLIDE, OPEN HAT ECHO PROB, OPEN HAT ECHO SWING, OPEN HAT SPREAD, PAD ATTACK, PAD COLOR, PAD DECAY, PAD DETUNE, PAD POCKET DEPTH, PAD SPACE, PAD WARP, PERC POSITION, PERC SPREAD, PERC TENSION, PERC TUNE, RUMBLE POCKET SHAPE, RUMBLE TUNE, RUMBLE WIDTH, VOICE ECHO FDBK, VOICE POCKET DEPTH, VOICE TUNE)	Travels over the bars set
ECHO SPACE, NEXT SONG, SLOW ALL, PREDELAY, ECHO TIME, DUB TIME, REPEAT LENGTH	Menus: at once, on the next buffer
GO, DROP, ENERGY TO THE CENTRE, THROW, FREEZE, ROLL	Presses: at once, as long as held or until pressed again
SEND A, SEND B, VOLUMES	Live's own volumes and sends: never glide
Not on the E16: MODEL (an instrument's sound)	At once. What sounds releases on the old model; the keys still down play again on the new one (a held chord goes on)
Not on the E16: CHORD, a dropped sample, any other menu or switch	At once: CHORD from the next note; a sample replaces the old one, notes sounding included

WHEN SOMETHING GOES WRONG

WHAT HAPPENED	DO THIS
A runaway: feedback, a stuck note, a wrong sound in front of the crowd	The interface's output down first: the room goes quiet, REC keeps recording. Then PANIC, held in SF CONTROL's window: every SF plugin fades out over 5 ms and clears; let go, it comes back without a click.
A plugin header shows FAULT in red	It caught an invalid value or a runaway and reset itself; it carries on. Note what was playing.
The wrong song landed	Turn NEXT SONG (15) to the right one and press it: it lands on the next bar.
A knob seems dead	Another page or scene is up. Or its range in Live's MIDI Mappings is narrow.
The ONE drifts or stops	Live is the clock: stop and start Live, the ONE follows. Check its USB cable before the show, not during it.
The ONE doesn't follow Live's tempo	Start from Live, not the ONE: Live sends clock only while it plays. The ONE's header shows USB (stopped: Shift + Stop until it does). Sync on OXI ONE MKII (Port 1)'s output and on no other port. Hold Shift + Back for 2 s: the ONE's MIDI monitor shows the clock arriving while Live plays. Still nothing: Sync on Port 2's output instead, never both.

8 BEFORE THE SHOW

Run through this list at every soundcheck. Each line is a thing that once went wrong somewhere.

CHECKLIST

1. Every SF plugin from the same build: SF CONTROL shows BUS SHARED and KICK REPORTING, and the pump lands on the kick.
2. NEXT SONG and GO recall the song: every track on the same bar, and a recalled knob is yours the moment you turn it.
3. The interface's output down: the room silent, REC still recording.
4. The PA pair carries PA and nothing else; Cue reaches the headphones and not the PA.
5. A THROW and ROLL each held for three seconds: on for the whole hold, off on release. DROP held for three seconds: one press (the cut), not two.
6. The shared E16 knobs don't jump (chapter 4); a PERF track page selects its track, NRG leaves it.
7. SLOW ALL where you want it for the opening (knob 1 pressed; OFF for direct moves), then back to the page's own knob.
8. A drop rehearsed with the night's TAIL and DROP DUCK settings: DROP, the tails, DROP again on the one.
9. A minute recorded on REC with a deep ROOM EQ cut: the cut isn't in the recording.
10. Soundcheck the busiest passage: SF LIMIT INPUT for 1-3 dB of limiting there, AUTO on, DOWN ONLY.
11. The room mics on ROOM: IN L and IN R move on its SF CAPTURE, the same gain on both, well under the top at the loudest.

CAPTURING THE ROOM

SF TERROIR can take the place you play in as a PLACE, without leaving Live: SF CAPTURE on ROOM plays the sweep to a speaker on the headphone out and records the room mics, and TERROIR turns the take into a place. SF TERROIR has to be in the set (on B SPACE in place of SF REVERB: controllers/e16 README, SF TERROIR ON B SPACE). At soundcheck or after the set: the transport stopped.

1. A speaker on the headphone out (a MiniRig on its line in), where the sound should come from, aimed at a wall or into the room. In TotalMix the mics must not reach the headphone out, or the speaker plays them and feeds back.
2. The mics where the audience will be, or where the place should be heard from (SF TERROIR's IN THE FIELD, MICS). The speaker as loud as it goes without buzzing; the headphone level sets it, SWEEP LEVEL on SF CAPTURE is only for headroom.
3. Stop the transport. In SF TERROIR click **CAPTURE** (amber), then the slot to fill. 46 s: keep the room quiet while the status line counts.
4. TERROIR analyses the take: the slot takes the name ROOM and the date and time; the status line should read SWEEP x3 and 30 dB or more over the noise. Under 30 dB: the speaker louder or the mics closer; THE MICS CLIPPED: the speaker or the gain down. Then again. THE SWEEP DIDN'T COME BACK: nothing reached the mics (TROUBLESHOOTING); the slot stays as it was.
5. PLACE is stored per song, so a recall goes back to the slot the song was stored with: STORE the songs that should play the new place. The take stays in Music/SF Live Suite/TERROIR/Captures.

CAPTURE again (it reads STOP) ends a take early; so do starting the transport, PANIC and Live's audio restarting, even before the sweep has started or while the take is being written. The sweep fades out in 10 ms and nothing is stored. The

PA stays silent: ROOM goes to the headphone out only.

RECORDING THE SET

— Press RESET on SF LIMIT, arm the instrument tracks, REC and ROOM, Automation Arm on, Arrangement record, play. The stereo recording on REC is the set, the one on ROOM the room mics for the mix; the MIDI and automation are for editing and rehearsal.

— Collect All and Save after the show. SF samples are inside the set; DECK tracks are referenced by path, so keep the files where they are.

TROUBLESHOOTING

SYMPTOM	CHECK
An instrument doesn't play	MIDI From: its ONE port and channel, never All Ins. The ONE's sequencer output matches: port A, the track's number (3 OPEN HAT on 3A).
Every instrument plays every part	A track's MIDI From is All Ins or All Channels.
Nothing ducks	SF CONTROL: BUS SHARED and KICK REPORTING. The kick must be SF KICK.
Pump early or late	Live buffer 512 or less; then POCKET OFFSET by ear. (The ONE's clock moves every part alike: not this.)
The drop comes in before the kick	SF BUILD: LAND ON KICK, with SF KICK playing before the cut (where its kicks land sets the drop: until four agree, Live's beat).
Songs don't change	Turn NEXT SONG (15), then press it (GO): SF CONTROL shows NEXT in amber, PRESS GO, until you do. The song must be stored (SONG nn EMPTY otherwise). SF CONTROL from this build: an older one has no NEXT SONG.
E16 rings don't follow	Remote on for E16 Port 1's output.
An E16 press does nothing	A scene file from before Nov 8 is loaded: load the current PERF and MIX. A toggle that flips back: Remote is on for two E16 ports.
No green on a travelling knob	SF CONTROL shows E16 RINGS PORT 3 in green? If not, the E16 isn't found: try PORT 1. Load the scenes from the repository again (the script is in them). Use the VST3s.
Knob 1 swapped (green) changes nothing	SLOW ALL isn't mapped in this set: Cmd-M, click SLOW ALL on SF CONTROL, press knob 1 and turn it. Cmd-M off, save.
Knob 1 moves its own parameter and SLOW ALL together	Its push was changed in the App: load PERF.oxie16 again (knob 1's push must be Swap Destination).
Every move glides (or none does)	SLOW ALL is saved with the set: press knob 1 and set it. A plugin whose own SLOW isn't CONTROL keeps its own.
An E16 knob jumps when touched	Auto Transmit is on, or the E16 keeps a value per page: chapter 4, check 3.
CONTROL shows LOCAL ONLY	The plugins come from different builds: rebuild and install the whole suite, restart Live.
A return doubles the dry signal	DRY off on SF DELAY or SF REVERB on the returns.
SF TERROIR won't capture	Its status line says why: the transport playing; NO SF CAPTURE (ROOM's device off); a take already running.
A capture says THE SWEEP DIDN'T COME BACK, or under 30 dB	The sweep didn't reach the speaker or the mics: the speaker on the headphone out, ROOM on and up; Input Config 1/2 a stereo pair, Monitor In. Then louder or closer. The slot stays as it was.
A capture says NO SWEEP IN THE RIGHT CHANNEL	That mic heard nothing (its cable, its gain): the place is mono. Fix it, capture again.
A capture says SF CAPTURE STOPPED ANSWERING	ROOM's SF CAPTURE was removed or switched off. Put it back, capture again: nothing was stored.

APPENDIX

E16 ADDRESS MAP (PERF, MIX)

Every address of the two scenes once, by channel: the Live target, the mapping, and where the knob is on the E16 (a PERF page is named by its track's code; LEVELS, SPACE and MACROS are MIX's). The same list is controllers/e16/sf-show-map.csv.

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
1	CC 12	1 KICK > SF KICK > TUNE	absolute	KCK 1
1	CC 13	1 KICK > SF KICK > PUNCH	absolute	KCK 2
1	CC 14	1 KICK > SF KICK > CLICK	absolute	KCK 3
1	CC 15	1 KICK > SF KICK > DRIVE	absolute	KCK 4
1	CC 16	1 KICK > SF KICK > PITCH TIME	absolute	KCK 5
1	CC 17	1 KICK > SF KICK > LAYER	absolute	KCK 6
1	CC 18	1 KICK > SF KICK > ECHO SPACE	absolute	KCK 7
1	CC 19	1 KICK > SF KICK > ECHO SWING	absolute	KCK 8
1	CC 20	1 KICK > SF KICK > DECAY	absolute	KCK 9
1	CC 21	1 KICK > SF KICK > DENSITY	absolute	KCK 10
1	CC 22	1 KICK > SF KICK > FILTER	absolute	KCK 11
1	CC 23	1 KICK > SF KICK > LEVEL	absolute	KCK 12
1	CC 28	1 KICK > SF ENERGY > ENERGY	absolute	NRG 1
1	CC 102	1 KICK > mixer > volume	absolute, Max 0 dB	LEVELS 1
1	CC 103	1 KICK > mixer > send A	absolute, Max 0 dB	KCK 13
1	CC 104	1 KICK > mixer > send B	absolute, Max 0 dB	KCK 14
2	CC 12	2 CLOSED HAT > SF PERC > TUNE	absolute	CHH 1
2	CC 13	2 CLOSED HAT > SF PERC > TONE	absolute	CHH 2
2	CC 14	2 CLOSED HAT > SF PERC > ACCENT	absolute	CHH 3
2	CC 15	2 CLOSED HAT > SF PERC > DRIVE	absolute	CHH 4
2	CC 16	2 CLOSED HAT > SF PERC > ECHO SPACE	absolute	CHH 5
2	CC 17	2 CLOSED HAT > SF PERC > ECHO SWING	absolute	CHH 6
2	CC 18	2 CLOSED HAT > SF PERC > ECHO PROB	absolute	CHH 7
2	CC 19	2 CLOSED HAT > SF PERC > SPREAD	absolute	CHH 8
2	CC 20	2 CLOSED HAT > SF PERC > DECAY	absolute	CHH 9
2	CC 21	2 CLOSED HAT > SF PERC > DENSITY	absolute	CHH 10
2	CC 22	2 CLOSED HAT > SF PERC > FILTER	absolute	CHH 11
2	CC 23	2 CLOSED HAT > SF PERC > LEVEL	absolute	CHH 12
2	CC 28	2 CLOSED HAT > SF ENERGY > ENERGY	absolute	NRG 2
2	CC 102	2 CLOSED HAT > mixer > volume	absolute, Max 0 dB	LEVELS 2
2	CC 103	2 CLOSED HAT > mixer > send A	absolute, Max 0 dB	CHH 13
2	CC 104	2 CLOSED HAT > mixer > send B	absolute, Max 0 dB	CHH 14
3	CC 12	3 OPEN HAT > SF PERC > TUNE	absolute	OHH 1
3	CC 13	3 OPEN HAT > SF PERC > TONE	absolute	OHH 2
3	CC 14	3 OPEN HAT > SF PERC > ACCENT	absolute	OHH 3

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
3	CC 15	3 OPEN HAT > SF PERC > DRIVE	absolute	OHH 4
3	CC 16	3 OPEN HAT > SF PERC > ECHO SPACE	absolute	OHH 5
3	CC 17	3 OPEN HAT > SF PERC > ECHO SWING	absolute	OHH 6
3	CC 18	3 OPEN HAT > SF PERC > ECHO PROB	absolute	OHH 7
3	CC 19	3 OPEN HAT > SF PERC > SPREAD	absolute	OHH 8
3	CC 20	3 OPEN HAT > SF PERC > DECAY	absolute	OHH 9
3	CC 21	3 OPEN HAT > SF PERC > DENSITY	absolute	OHH 10
3	CC 22	3 OPEN HAT > SF PERC > FILTER	absolute	OHH 11
3	CC 23	3 OPEN HAT > SF PERC > LEVEL	absolute	OHH 12
3	CC 28	3 OPEN HAT > SF ENERGY > ENERGY	absolute	NRG 3
3	CC 102	3 OPEN HAT > mixer > volume	absolute, Max 0 dB	LEVELS 3
3	CC 103	3 OPEN HAT > mixer > send A	absolute, Max 0 dB	OHH 13
3	CC 104	3 OPEN HAT > mixer > send B	absolute, Max 0 dB	OHH 14
4	CC 12	4 CLAP > SF PERC > TUNE	absolute	CLP 1
4	CC 13	4 CLAP > SF PERC > TONE	absolute	CLP 2
4	CC 14	4 CLAP > SF PERC > ACCENT	absolute	CLP 3
4	CC 15	4 CLAP > SF PERC > DRIVE	absolute	CLP 4
4	CC 16	4 CLAP > SF PERC > ECHO SPACE	absolute	CLP 5
4	CC 17	4 CLAP > SF PERC > ECHO SWING	absolute	CLP 6
4	CC 18	4 CLAP > SF PERC > ECHO PROB	absolute	CLP 7
4	CC 19	4 CLAP > SF PERC > SPREAD	absolute	CLP 8
4	CC 20	4 CLAP > SF PERC > DECAY	absolute	CLP 9
4	CC 21	4 CLAP > SF PERC > DENSITY	absolute	CLP 10
4	CC 22	4 CLAP > SF PERC > FILTER	absolute	CLP 11
4	CC 23	4 CLAP > SF PERC > LEVEL	absolute	CLP 12
4	CC 28	4 CLAP > SF ENERGY > ENERGY	absolute	NRG 4
4	CC 102	4 CLAP > mixer > volume	absolute, Max 0 dB	LEVELS 4
4	CC 103	4 CLAP > mixer > send A	absolute, Max 0 dB	CLP 13
4	CC 104	4 CLAP > mixer > send B	absolute, Max 0 dB	CLP 14
5	CC 12	5 PERC > SF STRIKE > POWER	absolute	PRC 1
5	CC 13	5 PERC > SF STRIKE > DISTANCE	absolute	PRC 2
5	CC 14	5 PERC > SF STRIKE > SIZE	absolute	PRC 3
5	CC 15	5 PERC > SF STRIKE > MATERIAL	absolute	PRC 4
5	CC 16	5 PERC > SF STRIKE > POSITION	absolute	PRC 5
5	CC 17	5 PERC > SF STRIKE > TENSION	absolute	PRC 6
5	CC 18	5 PERC > SF STRIKE > SPREAD	absolute	PRC 7
5	CC 19	5 PERC > SF STRIKE > TUNE	absolute	PRC 8
5	CC 20	5 PERC > SF STRIKE > DECAY	absolute	PRC 9
5	CC 21	5 PERC > SF STRIKE > DENSITY	absolute	PRC 10
5	CC 22	5 PERC > SF STRIKE > FILTER	absolute	PRC 11
5	CC 23	5 PERC > SF STRIKE > LEVEL	absolute	PRC 12
5	CC 28	5 PERC > SF ENERGY > ENERGY	absolute	NRG 5
5	CC 102	5 PERC > mixer > volume	absolute, Max 0 dB	LEVELS 5

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
5	CC 103	5 PERC > mixer > send A	absolute, Max 0 dB	PRC 13
5	CC 104	5 PERC > mixer > send B	absolute, Max 0 dB	PRC 14
6	CC 12	6 BASS > SF BASS > CUTOFF	absolute	BAS 1
6	CC 13	6 BASS > SF BASS > RES	absolute	BAS 2
6	CC 14	6 BASS > SF BASS > ENV AMT	absolute	BAS 3
6	CC 15	6 BASS > SF BASS > DECAY	absolute	BAS 4
6	CC 16	6 BASS > SF BASS > DRIVE	absolute	BAS 5
6	CC 17	6 BASS > SF BASS > ACCENT	absolute	BAS 6
6	CC 18	6 BASS > SF BASS > GLIDE	absolute	BAS 7
6	CC 19	6 BASS > SF BASS > POCKET DEPTH	absolute	BAS 8
6	CC 20	6 BASS > SF BASS > GATE	absolute	BAS 9
6	CC 21	6 BASS > SF BASS > DENSITY	absolute	BAS 10
6	CC 22	6 BASS > SF BASS > FILTER	absolute	BAS 11
6	CC 23	6 BASS > SF BASS > LEVEL	absolute	BAS 12
6	CC 28	6 BASS > SF ENERGY > ENERGY	absolute	NRG 6
6	CC 102	6 BASS > mixer > volume	absolute, Max 0 dB	LEVELS 6
6	CC 103	6 BASS > mixer > send A	absolute, Max 0 dB	BAS 13
6	CC 104	6 BASS > mixer > send B	absolute, Max 0 dB	BAS 14
7	CC 12	7 RUMBLE > SF RUMBLE > SIZE	absolute	RMB 1
7	CC 13	7 RUMBLE > SF RUMBLE > DARK	absolute	RMB 2
7	CC 14	7 RUMBLE > SF RUMBLE > GRIT	absolute	RMB 3
7	CC 15	7 RUMBLE > SF RUMBLE > PUMP	absolute	RMB 4
7	CC 16	7 RUMBLE > SF RUMBLE > TUNE	absolute	RMB 5
7	CC 17	7 RUMBLE > SF RUMBLE > PREDELAY	absolute	RMB 6
7	CC 18	7 RUMBLE > SF RUMBLE > WIDTH	absolute	RMB 7
7	CC 19	7 RUMBLE > SF RUMBLE > POCKET SHAPE	absolute	RMB 8
7	CC 20	7 RUMBLE > SF RUMBLE > GATE	absolute	RMB 9
7	CC 21	7 RUMBLE > SF RUMBLE > DENSITY	absolute	RMB 10
7	CC 22	7 RUMBLE > SF RUMBLE > FILTER	absolute	RMB 11
7	CC 23	7 RUMBLE > SF RUMBLE > LEVEL	absolute	RMB 12
7	CC 28	7 RUMBLE > SF ENERGY > ENERGY	absolute	NRG 7
7	CC 102	7 RUMBLE > mixer > volume	absolute, Max 0 dB	LEVELS 7
7	CC 103	7 RUMBLE > mixer > send A	absolute, Max 0 dB	RMB 13
7	CC 104	7 RUMBLE > mixer > send B	absolute, Max 0 dB	RMB 14
8	CC 12	8 LEAD > SF LEAD > CUTOFF	absolute	LED 1
8	CC 13	8 LEAD > SF LEAD > RES	absolute	LED 2
8	CC 14	8 LEAD > SF LEAD > ENV AMT	absolute	LED 3
8	CC 15	8 LEAD > SF LEAD > DECAY	absolute	LED 4
8	CC 16	8 LEAD > SF LEAD > DRIVE	absolute	LED 5
8	CC 17	8 LEAD > SF LEAD > ACCENT	absolute	LED 6
8	CC 18	8 LEAD > SF LEAD > GLIDE	absolute	LED 7
8	CC 19	8 LEAD > SF LEAD > COLOR	absolute	LED 8
8	CC 20	8 LEAD > SF LEAD > GATE	absolute	LED 9

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
8	CC 21	8 LEAD > SF LEAD > DENSITY	absolute	LED 10
8	CC 22	8 LEAD > SF LEAD > FILTER	absolute	LED 11
8	CC 23	8 LEAD > SF LEAD > LEVEL	absolute	LED 12
8	CC 28	8 LEAD > SF ENERGY > ENERGY	absolute	NRG 8
8	CC 29	8 LEAD > SF LEAD > ACID	absolute	NRG 12
8	CC 102	8 LEAD > mixer > volume	absolute, Max 0 dB	LEVELS 8
8	CC 103	8 LEAD > mixer > send A	absolute, Max 0 dB	LED 13
8	CC 104	8 LEAD > mixer > send B	absolute, Max 0 dB	LED 14
9	CC 12	9 CHORD > SF LEAD > CUTOFF	absolute	CRD 1
9	CC 13	9 CHORD > SF LEAD > RES	absolute	CRD 2
9	CC 14	9 CHORD > SF LEAD > ENV AMT	absolute	CRD 3
9	CC 15	9 CHORD > SF LEAD > DECAY	absolute	CRD 4
9	CC 16	9 CHORD > SF LEAD > ATTACK	absolute	CRD 5
9	CC 17	9 CHORD > SF LEAD > SPACE	absolute	CRD 6
9	CC 18	9 CHORD > SF LEAD > DETUNE	absolute	CRD 7
9	CC 19	9 CHORD > SF LEAD > POCKET DEPTH	absolute	CRD 8
9	CC 20	9 CHORD > SF LEAD > GATE	absolute	CRD 9
9	CC 21	9 CHORD > SF LEAD > DENSITY	absolute	CRD 10
9	CC 22	9 CHORD > SF LEAD > FILTER	absolute	CRD 11
9	CC 23	9 CHORD > SF LEAD > LEVEL	absolute	CRD 12
9	CC 28	9 CHORD > SF ENERGY > ENERGY	absolute	NRG 9
9	CC 102	9 CHORD > mixer > volume	absolute, Max 0 dB	LEVELS 9
9	CC 103	9 CHORD > mixer > send A	absolute, Max 0 dB	CRD 13
9	CC 104	9 CHORD > mixer > send B	absolute, Max 0 dB	CRD 14
10	CC 12	10 PAD > SF PAD > CUTOFF	absolute	PAD 1
10	CC 13	10 PAD > SF PAD > RES	absolute	PAD 2
10	CC 14	10 PAD > SF PAD > COLOR	absolute	PAD 3
10	CC 15	10 PAD > SF PAD > DETUNE	absolute	PAD 4
10	CC 16	10 PAD > SF PAD > ATTACK	absolute	PAD 5
10	CC 17	10 PAD > SF PAD > DECAY	absolute	PAD 6
10	CC 18	10 PAD > SF PAD > SPACE	absolute	PAD 7
10	CC 19	10 PAD > SF PAD > POCKET DEPTH	absolute	PAD 8
10	CC 20	10 PAD > SF PAD > GATE	absolute	PAD 9
10	CC 21	10 PAD > SF PAD > DENSITY	absolute	PAD 10
10	CC 22	10 PAD > SF PAD > FILTER	absolute	PAD 11
10	CC 23	10 PAD > SF PAD > LEVEL	absolute	PAD 12
10	CC 28	10 PAD > SF ENERGY > ENERGY	absolute	NRG 10
10	CC 29	10 PAD > SF PAD > WARP	absolute	NRG 14
10	CC 102	10 PAD > mixer > volume	absolute, Max 0 dB	LEVELS 10
10	CC 103	10 PAD > mixer > send A	absolute, Max 0 dB	PAD 13
10	CC 104	10 PAD > mixer > send B	absolute, Max 0 dB	PAD 14
11	CC 12	11 VOICE > SF VOICE > SIGNAL	absolute	VOX 1
11	CC 13	11 VOICE > SF VOICE > TONE	absolute	VOX 2

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
11	CC 14	11 VOICE > SF VOICE > TUNE	absolute	VOX 3
11	CC 15	11 VOICE > SF VOICE > DRIVE	absolute	VOX 4
11	CC 16	11 VOICE > SF VOICE > ECHO	absolute	VOX 5
11	CC 17	11 VOICE > SF VOICE > ECHO TIME	absolute	VOX 6
11	CC 18	11 VOICE > SF VOICE > ECHO FDBK	absolute	VOX 7
11	CC 19	11 VOICE > SF VOICE > POCKET DEPTH	absolute	VOX 8
11	CC 20	11 VOICE > SF VOICE > GATE	absolute	VOX 9
11	CC 21	11 VOICE > SF VOICE > DENSITY	absolute	VOX 10
11	CC 22	11 VOICE > SF VOICE > FILTER	absolute	VOX 11
11	CC 23	11 VOICE > SF VOICE > LEVEL	absolute	VOX 12
11	CC 28	11 VOICE > SF ENERGY > ENERGY	absolute	NRG 11
11	CC 102	11 VOICE > mixer > volume	absolute, Max 0 dB	LEVELS 11
11	CC 103	11 VOICE > mixer > send A	absolute, Max 0 dB	VOX 13
11	CC 104	11 VOICE > mixer > send B	absolute, Max 0 dB	VOX 14
12	CC 44	A DUB > SF DELAY > TIME	absolute	SPACE 1
12	CC 45	A DUB > SF DELAY > FEEDBACK	absolute, 0-70 % at first	SPACE 2
12	CC 46	A DUB > SF DELAY > TONE	absolute	SPACE 3
12	CC 47	A DUB > SF DELAY > DRIVE	absolute	SPACE 4
12	CC 60	A DUB > SF DELAY > THROW	press: hold	SPACE 1
12	CC 102	A DUB > mixer > volume	absolute, Max 0 dB	LEVELS 15
12	NOTE 61 (C#3)	A DUB > SF DELAY > FREEZE	press: toggle	SPACE 2
13	CC 48	B SPACE > SF REVERB > SIZE	absolute	SPACE 5
13	CC 49	B SPACE > SF REVERB > DECAY	absolute	SPACE 6
13	CC 50	B SPACE > SF REVERB > DARK	absolute	SPACE 7
13	CC 51	B SPACE > SF REVERB > DUCK	absolute	SPACE 8
13	CC 64	B SPACE > SF REVERB > THROW	press: hold	SPACE 5
13	CC 102	B SPACE > mixer > volume	absolute, Max 0 dB	LEVELS 16
13	NOTE 65 (F3)	B SPACE > SF REVERB > FREEZE	press: toggle	SPACE 6
14	CC 52	DRUMS > SF REPEAT > LENGTH	absolute	MACROS 9
14	CC 53	DRUMS > SF REPEAT > PITCH	absolute	MACROS 10
14	CC 68	DRUMS > SF REPEAT > ROLL	press: hold	MACROS 9
14	CC 102	DRUMS > mixer > volume	absolute, Max 0 dB	LEVELS 12
14	CC 103	LOW > mixer > volume	absolute, Max 0 dB	LEVELS 13
14	CC 104	MUSIC > mixer > volume	absolute, Max 0 dB	LEVELS 14
15	CC 27	LIVE BUS > SF BUILD > BUILD	absolute	every PERF page, encoder 16
15	CC 29	LIVE BUS > SF DRIVE > MIX	absolute	NRG 13
15	CC 43	LIVE BUS > SF BUILD > DROP	press: hold	every PERF page, encoder 16
16	CC 12	CONTROL > SF CONTROL > SLOW ALL	absolute	every PERF page, encoder 1 (second destination), MACROS 7
16	CC 26	CONTROL > SF CONTROL > NEXT SONG	absolute	every PERF page, encoder 15

CH	MESSAGE	LIVE TARGET	MAPPING	ON THE E16
16	CC 42	CONTROL > SF CONTROL > GO	press: hold	every PERF page, encoder 15
16	CC 44	CONTROL > SF CONTROL > MACRO 1	absolute	MACROS 1
16	CC 45	CONTROL > SF CONTROL > MACRO 2	absolute	MACROS 2
16	CC 46	CONTROL > SF CONTROL > MACRO 3	absolute	MACROS 3
16	CC 47	CONTROL > SF CONTROL > MACRO 4	absolute	MACROS 4
16	CC 48	CONTROL > SF CONTROL > MACRO 5	absolute	MACROS 5
16	CC 49	CONTROL > SF CONTROL > MACRO 6	absolute	MACROS 6